2
The Two-Body Problem

So we grew together,
Like to a double cherry, seeming parted,
But yet an union in partition –
Two lovely berries moulded on one stem;
So, with two seeming bodies, but one heart.

William Shakespeare, *A Midsummer Night's Dream, II, ii*

2.1 Introduction

The two-body problem is perhaps the simplest, integrable problem in solar system dynamics. It concerns the interaction of two point masses moving under a mutual gravitational attraction described by Newton's universal law of gravitation, Eq. (1.1). The wide variety of masses in the solar system permits the orbits of most planets and satellites to be approximated by two-body motion, consisting of a smaller body moving around a much larger central body. The effects of other bodies can usually be treated as perturbations to the two-body system. For example, the path of Jupiter (mass $m_{\mathrm{J}} = 1.9 \times 10^{27}$ kg) around the Sun (mass $m_{\mathrm{Su}} = 2.0 \times 10^{30}$ kg $= 1000\, m_{\mathrm{J}}$) is basically an ellipse with the principal perturbations coming from the other planets, notably Saturn (mass $m_{\mathrm{Sa}} = 5.7 \times 10^{26}$ kg).

In Volume I of his *Principia*, Isaac Newton (1687) showed that only two types of central force could give rise to the observed elliptical motion of the planets. The first was a linear force directed towards the centre of the ellipse, and the second was an inverse square force directed towards one focus of the ellipse. However, only the second type of force can give rise to Kepler's empirical laws of planetary motion (Sect. 1.3). In this chapter we derive the basic equations of planetary motion and solve the two-body problem, showing how Kepler's laws arise.

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2.2 Equations of Motion

Consider the motion of two masses m_1 and m_2 with position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ referred to some origin O fixed in inertial space (see Fig. 2.1).

The vector $\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$ denotes the relative position of the mass m_2 with respect to m_1. The gravitational forces and the consequent accelerations experienced by the two masses are

$$\mathbf{F}_1 = +\mathcal{G}\frac{m_1 m_2}{r^3}\mathbf{r} = m_1\ddot{\mathbf{r}}_1 \qquad \text{and} \qquad \mathbf{F}_2 = -\mathcal{G}\frac{m_1 m_2}{r^3}\mathbf{r} = m_2\ddot{\mathbf{r}}_2 \tag{2.1}$$

respectively, where $\mathcal{G} = 6.67260 \times 10^{-11}\mathrm{Nm}^2\mathrm{kg}^{-2}$ is the *universal gravitational constant*. Thus

$$m_1\ddot{\mathbf{r}}_1 + m_2\ddot{\mathbf{r}}_2 = 0, \tag{2.2}$$

which can be integrated directly twice to give

$$m_1\dot{\mathbf{r}}_1 + m_2\dot{\mathbf{r}}_2 = \mathbf{a} \qquad \text{and} \qquad m_1\mathbf{r}_1 + m_2\mathbf{r}_2 = \mathbf{a}t + \mathbf{b}, \tag{2.3}$$

where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors. If $\mathbf{R} = (m_1\mathbf{r}_1 + m_2\mathbf{r}_2)/(m_1 + m_2)$ denotes the position vector of the centre of mass, then Eqs. (2.3) can be written

$$\dot{\mathbf{R}} = \frac{\mathbf{a}}{m_1 + m_2} \qquad \text{and} \qquad \mathbf{R} = \frac{\mathbf{a}t + \mathbf{b}}{m_1 + m_2}. \tag{2.4}$$

This implies that either the centre of mass is stationary (if $\mathbf{a} = 0$) or it is moving with a constant velocity in a straight line with respect to the origin O. Note that this result is not specific to the inverse square law of force.

Now consider the motion of m_2 with respect to m_1. This allows us to simplify the problem without losing any of its essential features. In Sect. 2.7 we shall revert to considering motion in the centre of mass frame. Writing $\ddot{\mathbf{r}} = \ddot{\mathbf{r}}_2 - \ddot{\mathbf{r}}_1$, and using Eq. (2.1), we obtain

$$\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} + \mu\frac{\mathbf{r}}{r^3} = 0, \tag{2.5}$$

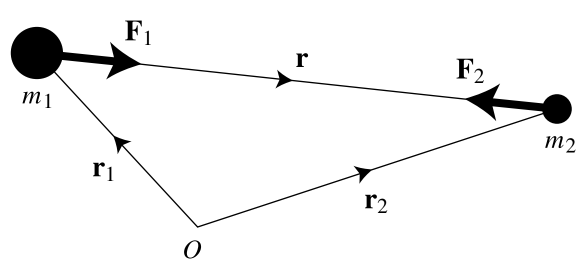

Fig. 2.1. A vector diagram for the forces acting on two masses, m_1 and m_2, with position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$.

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where $\mu = \mathcal{G}(m_1 + m_2)$. This is the *equation of relative motion*. In order to solve it and find the path of m_2 relative to m_1 we must first derive several constants of the motion.

Taking the vector product of $\mathbf{r}$ with Eq. (2.5) we have $\mathbf{r} \times \ddot{\mathbf{r}} = 0$, which can be integrated directly to give

$$\mathbf{r} \times \dot{\mathbf{r}} = \mathbf{h}, \tag{2.6}$$

where $\mathbf{h}$ is a constant vector perpendicular to both $\mathbf{r}$ and $\dot{\mathbf{r}}$. Hence the motion of m_2 about m_1 lies in a plane perpendicular to the direction defined by $\mathbf{h}$. This also implies that the position and velocity vectors always lie in the same plane (see Fig. 2.2). Equation (2.6) is commonly referred to as the *angular momentum integral*. However, although $h = |\mathbf{h}|$ for systems in which $m_2 \ll m_1$ is approximately equal to the orbital angular momentum per unit mass of the body m_2, it is not the actual angular momentum in the inertial system since this is calculated using the position and velocity vectors referred to the centre of mass. We consider this in more detail in Sect. 2.7.

Since $\mathbf{r}$ and $\dot{\mathbf{r}}$ always lie in the same plane (the *orbit plane*) it is natural that we now restrict ourselves to considering motion in that plane; the motion referred to an arbitrary reference frame is considered in Sect. 2.8. We now transform to a polar coordinate system (r, θ) referred to an origin centred on the mass m_1 and an arbitrary reference line corresponding to $\theta = 0$. Note that even though the centre of mass of m_1 and m_2 could be moving in inertial space, the direction of the reference line remains fixed. If we let $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$ denote unit vectors along and perpendicular to the radius vector respectively, then the position, velocity, and acceleration vectors can be written in polar coordinates as

$$\mathbf{r} = r\hat{\mathbf{r}}, \qquad \dot{\mathbf{r}} = \dot{r}\hat{\mathbf{r}} + r\dot{\theta}\hat{\boldsymbol{\theta}}, \qquad \ddot{\mathbf{r}} = (\ddot{r} - r\dot{\theta}^2)\hat{\mathbf{r}} + \left[\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}t}\left(r^2\dot{\theta}\right)\right]\hat{\boldsymbol{\theta}}. \tag{2.7}$$

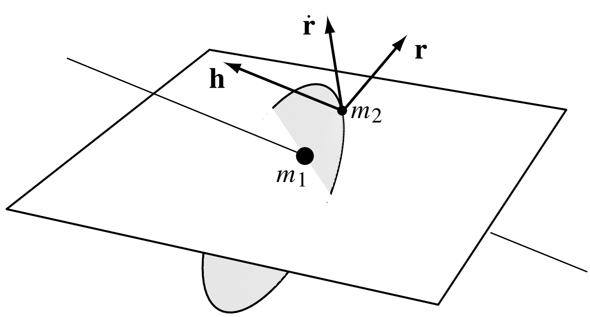

Fig. 2.2. The motion of m_2 with respect to m_1 defines an orbital plane (shaded region), because $\mathbf{r} \times \dot{\mathbf{r}}$ is a constant vector, $\mathbf{h}$, the angular momentum vector, and this is always perpendicular to the orbit plane.

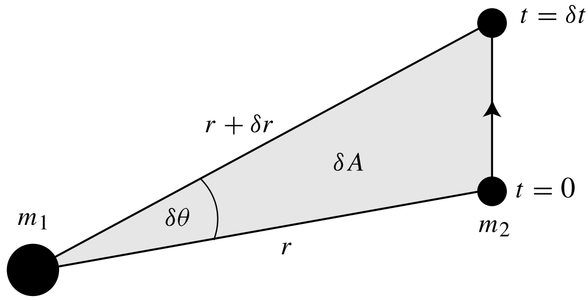

Fig. 2.3. The area δA swept out in a time δt as a position vector moves through an angle $\delta\theta$.

Substituting the expression for $\dot{\mathbf{r}}$ into Eq. (2.6) gives $\mathbf{h} = r^2\dot{\theta}\hat{\mathbf{z}}$, where $\hat{\mathbf{z}}$ is a unit vector perpendicular to the plane of the orbit forming a right-handed triad with $\hat{\mathbf{r}}$ and $\hat{\boldsymbol{\theta}}$. Hence

$$h = r^2\dot{\theta}\,. \tag{2.8}$$

Consider the motion of the mass m_2 during a time interval δt (see Fig. 2.3). At time $t = 0$ it has polar coordinates $(r,\ \theta)$, while at time $t + \delta t$ its polar coordinates have changed to $(r + \delta r,\ \theta + \delta\theta)$. The area swept out by the radius vector in time δt is

$$\delta A \approx \frac{1}{2} r\,(r + \delta r) \sin\delta\theta \approx \frac{1}{2} r^2 \delta\theta, \tag{2.9}$$

where we have neglected second- and higher-order terms in the small quantities. Hence, by dividing each side by δt and taking the limit as $\delta t \to 0$ we have

$$\frac{\mathrm{d}A}{\mathrm{d}t} = \frac{1}{2} r^2 \frac{\mathrm{d}\theta}{\mathrm{d}t} = \frac{1}{2} h. \tag{2.10}$$

Since h is a constant this implies that equal areas are swept out in equal times and hence Eq. (2.10) is the mathematical form of Kepler's second law of planetary motion (see Sect. 1.3). Note that this does not require an inverse square law of force, but only that the force is directed along the line joining the two masses.

2.3 Orbital Position and Velocity

We obtain a scalar equation for the relative motion by substituting the expression for $\ddot{\mathbf{r}}$ from Eq. (2.7) into Eq. (2.5); comparing the $\hat{\mathbf{r}}$ components gives

$$\ddot{r} - r\dot{\theta}^2 = -\frac{\mu}{r^2}\,. \tag{2.11}$$

To solve this equation and find r as a function of θ we need to make the substitution $u = 1/r$ and to eliminate the time by making use of the constant $h = r^2\dot{\theta}$. By

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differentiating r with respect to time, we obtain

$$\dot{r} = -\frac{1}{u^2}\frac{\mathrm{d}u}{\mathrm{d}\theta}\dot{\theta} = -h\frac{\mathrm{d}u}{\mathrm{d}\theta} \qquad \text{and} \qquad \ddot{r} = -h\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}\dot{\theta} = -h^2u^2\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} \tag{2.12}$$

and hence Eq. (2.11) can be written

$$\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} + u = \frac{\mu}{h^2}\,. \tag{2.13}$$

This is a second-order, linear differential equation with a general solution

$$u = \frac{\mu}{h^2}\left[1 + e\cos(\theta - \varpi)\right], \tag{2.14}$$

where e (an amplitude) and ϖ (a phase) are two constants of integration. Substituting back for r we have

$$r = \frac{p}{1 + e\cos(\theta - \varpi)}, \tag{2.15}$$

which is the general equation of a conic in polar coordinates where e is the *eccentricity* and p is the *semilatus rectum* given by

$$p = h^2/\mu\,. \tag{2.16}$$

The four possible conics are:

$$\begin{array}{lll} \text{circle:} & e = 0, & p = a; \\ \text{ellipse:} & 0 < e < 1, & p = a(1 - e^2); \\ \text{parabola:} & e = 1, & p = 2q; \\ \text{hyperbola:} & e > 1, & p = a(e^2 - 1), \end{array} \tag{2.17}$$

where the constant a is the *semi-major axis* of the conic. In the special case of the parabola p is defined in terms of q, the distance to the central mass at closest approach. The conic section curves derive their name from the curves formed by the intersection of various planes with the surface of a cone (see Fig. 2.4).

The type of conic is determined by the angle the plane makes with the horizontal. If the plane is horizontal, that is, perpendicular to the axis of symmetry of the cone, then the resulting curve is a circle. If the angle is less than the slope angle of the cone then an ellipse results, whereas if the plane is parallel to the slope of the cone a parabola results. A hyperbola results if the angle the plane makes with the horizontal is anywhere between the slope angle of the cone and the vertical.

In the context of the two-body problem the path of a planet about the Sun is elliptical and closed in inertial space (see Fig. 2.5), and hence Kepler's first law of planetary motion (see Sect. 1.3) is a consequence of the inverse square law of force. Note that the mass m_1 fills one focus of the ellipse while the other focus is empty.

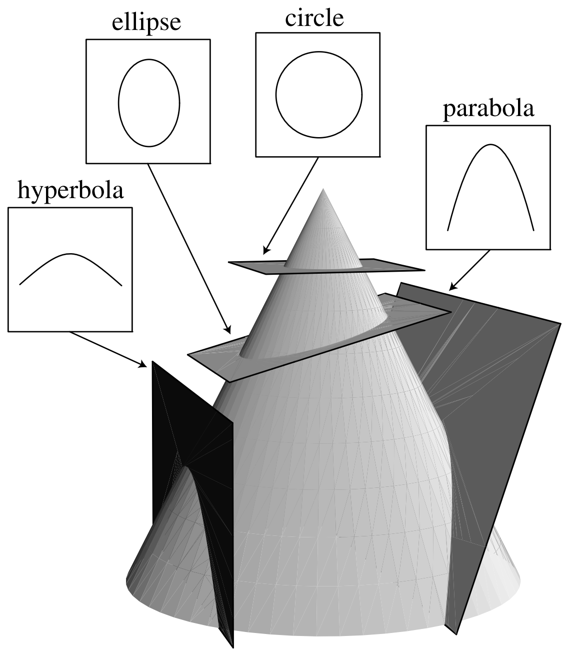

Fig. 2.4. The intersections of planes at different angles with the surface of a cone form the family of curves known as the conic sections.

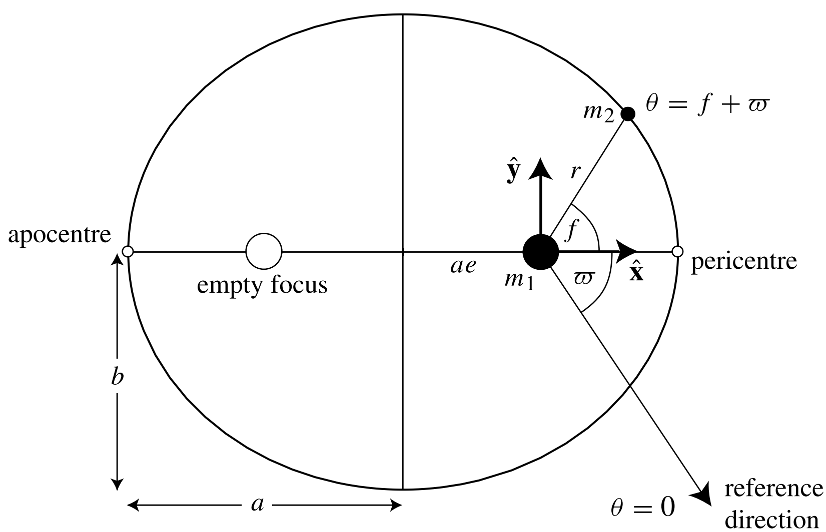

Fig. 2.5. The geometry of the ellipse of semi-major axis a, semi-minor axis b, eccentricity e, and longitude of pericentre ϖ.

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Although a large number of cometary orbits have $e \approx 1$, most permanent members of the solar system have $e \ll 1$. The notable exceptions among the planets are Pluto ($e = 0.25$) and Mercury ($e = 0.21$), while Nereid ($e = 0.75$), a moon of Neptune, has the largest eccentricity of any known natural satellite. Consequently, throughout most of this book we concentrate on elliptical motion. In this case $p = a(1 - e^2)$ and the quantities a and e are related by

$$b^2 = a^2(1 - e^2), \tag{2.18}$$

where b is the *semi-minor axis* of the ellipse (see Fig. 2.5); we also have

$$r = \frac{a(1 - e^2)}{1 + e\cos(\theta - \varpi)}. \tag{2.19}$$

In celestial mechanics it is customary to use the term longitude when referring to any angle that is measured with respect to a reference line fixed in inertial space. The angle θ is called the *true longitude*. A simple inspection of Eq. (2.19) shows that the minimum and maximum values of the orbital radius are $r_{\rm p} = a(1 - e)$ and $r_{\rm a} = a(1 + e)$, which occur when $\theta = \varpi$ and $\theta = \varpi + \pi$ respectively. These points in the orbit are called the *pericentre* (or *periapse*) and the *apocentre* (or *apoapse*) respectively, although other names can be used for particular systems (e.g., perihelion, perijove, periselenium). Note that the distance of either focus from the centre of the ellipse is ae.

The angle ϖ (pronounced "curly pi") is called the *longitude of pericentre*. Although this is a constant for the two-body problem, it can vary with time when additional perturbations are introduced (see Chaps. 3, 6–8). It is usually more convenient to refer the angular coordinate to the pericentre rather than to the arbitrary reference line. This leads to the introduction of the angle $f = \theta - \varpi$ (see Fig. 2.5), which is called the *true anomaly*. Since ϖ is constant the path is closed and the angular position is described by f or θ, which are 2π-periodic variables. Hence Eq. (2.19) can be written

$$r = \frac{a(1 - e^2)}{1 + e\cos f}. \tag{2.20}$$

Using a Cartesian coordinate system centred on the central mass with the x axis pointing towards the pericentre (see Fig. 2.5), the components of the position vector are

$$x = r\cos f \qquad \text{and} \qquad y = r\sin f. \tag{2.21}$$

In one orbital period T the area swept out by a radius vector is simply the area $A = \pi ab$ enclosed by the ellipse. From Eq. (2.10) this area has to equal $hT/2$ and hence, given that $h^2 = \mu a(1 - e^2)$,

$$T^2 = \frac{4\pi^2}{\mu}a^3, \tag{2.22}$$

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which corresponds to Kepler's third law of planetary motion (see Sect. 1.3). Note that the period of the orbit is independent of e and is a function of μ and a only.

Consider the case of two objects of mass m and m', orbiting a central object of mass m_c. Let the orbiting objects have semi-major axes a and a' and orbital periods T and T'. Equation (2.22) gives

$$\frac{m_c + m}{m_c + m'} = \left(\frac{a}{a'}\right)^3 \left(\frac{T'}{T}\right)^2 . \tag{2.23}$$

In the case of planets orbiting the Sun we have $m, m' \ll m_c$ and hence $(a/a')^3 \approx (T/T')^2$. Therefore, if a and T denote the values of the semi-major axis and the period of the Earth's orbit and the unit of length is taken to be the *astronomical unit AU* (1 AU is the approximate semi-major axis of the Earth's orbit) and the unit of time is taken to be the year (the approximate period of the Earth's orbit), we have $T' \approx a'^{3/2}$.

If any solar system object (e.g., an asteroid or a comet) has a small natural or artificial satellite, then observations of the distance and period of the satellite can be used with Kepler's third law to derive an estimate of the mass of the object. Consider Eq. (2.22) applied to the Sun–object and object–satellite systems. Let m_c, m, and m' now denote the masses of the Sun, object, and satellite respectively with similar definitions for the semi-major axes and orbital periods. This gives

$$\frac{m + m'}{m_c + m} \approx \frac{m}{m_c} = \left(\frac{a'}{a}\right)^3 \left(\frac{T}{T'}\right)^2 , \tag{2.24}$$

where we have taken $m' \ll m$ and $m \ll m_c$. This means that the mass of the object can be estimated from the orbital properties of its satellite.

Figure 2.6 shows an image of the asteroid (243) Ida and its moon Dactyl taken by the *Galileo* spacecraft on its way to Jupiter. Direct estimates of the masses of asteroids are notoriously difficult because of their small size (the largest asteroid, (1) Ceres, has a diameter of 913 km) and hence their small perturbations on other objects. Although analysis of the *Viking* data (Standish & Hellings 1989) has permitted mass determinations of the larger objects due to their direct perturbations on the orbit of Mars, similar calculations for the smaller asteroids are almost impossible. However, observations of Dactyl's motion using images obtained by the *Galileo* spacecraft have resulted in an estimated density of 2.6 ± 0.5 g cm^{-3} (Belton et al. 1995).

Since the angle θ covers 2π radians in one orbital period we can define the "average" angular velocity, or the *mean motion*, n as

$$n = \frac{2\pi}{T} \tag{2.25}$$

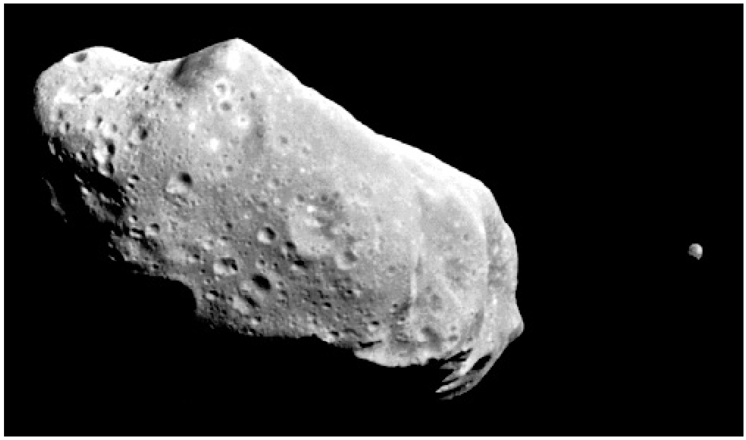

Fig. 2.6. An image of the asteroid (243) Ida and its moon Dactyl taken by the *Galileo* spacecraft on 28th August 1993. Ida is approximately $56 \times 24 \times 21$ km in size and its moon Dactyl is about 1.4 km across. *(Image courtesy of NASA/JPL.)*

and we can write

$$\mu = n^2 a^3 \quad \text{and} \quad h = na^2\sqrt{1-e^2} = \sqrt{\mu a(1-e^2)}\,. \tag{2.26}$$

Although the mean motion is constant in the two-body problem, the actual angular velocity $\dot{f}$ of the orbiting body *is* a function of the longitude.

We can derive another constant of the motion by taking the scalar product of $\dot{\mathbf{r}}$ with Eq. (2.5) and using the expressions for $\mathbf{r}$ and $\dot{\mathbf{r}}$ from Eq. (2.7). This gives the scalar equation

$$\dot{\mathbf{r}} \cdot \ddot{\mathbf{r}} + \mu \frac{\dot{r}}{r^2} = 0, \tag{2.27}$$

which can be integrated to give

$$\frac{1}{2}v^2 - \frac{\mu}{r} = C, \tag{2.28}$$

where $v^2 = \dot{\mathbf{r}} \cdot \dot{\mathbf{r}}$ is the square of the velocity and C is a constant of the motion. Equation (2.28), often called the *vis viva integral*, shows that the orbital energy per unit mass is conserved. Thus the two-body problem has four constants of the motion: the energy integral C and the three components of the angular momentum integral, $\mathbf{h}$. Note that it is also possible to express these constants in different forms such as the orbital elements, or quantities such as the eccentricity vector (see Question 2.4).

By finding another expression for v^2 we can derive an expression for C. Since ϖ is fixed we have $\dot{\theta} = \mathrm{d}(f + \varpi)/\mathrm{d}t = \dot{f}$ and using the definition of $\dot{\mathbf{r}}$ from Eq. (2.7) gives

$$v^2 = \dot{\mathbf{r}} \cdot \dot{\mathbf{r}} = \dot{r}^2 + r^2 \dot{f}^2\,. \tag{2.29}$$

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Differentiating Eq. (2.20) we have

$$\dot{r} = \frac{r \dot{f} e \sin f}{1 + e \cos f}. \tag{2.30}$$

Using $r^2 \dot{f} = h = na^2\sqrt{1-e^2}$, we can write

$$\dot{r} = \frac{na}{\sqrt{1-e^2}} e \sin f \tag{2.31}$$

and

$$r\dot{f} = \frac{na}{\sqrt{1-e^2}}(1 + e \cos f), \tag{2.32}$$

so that Eq. (2.29) can be written

$$v^2 = \frac{n^2a^2}{1-e^2}(1 + 2e\cos f + e^2) = \frac{n^2a^2}{1-e^2}\left(\frac{2a(1-e^2)}{r} - (1-e^2)\right). \tag{2.33}$$

Hence

$$v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right). \tag{2.34}$$

Consequently, the velocity of the orbiting body is a maximum at pericentre ($f = 0$) and a minimum at apocentre ($f = \pi$). The respective values are

$$v_\mathrm{p} = na\sqrt{\frac{1+e}{1-e}} \quad \text{and} \quad v_\mathrm{a} = na\sqrt{\frac{1-e}{1+e}}. \tag{2.35}$$

We can also find the x and y components of the velocity vector by taking the time derivatives of the expressions for x and y in Eq. (2.21) and substituting the expressions for $\dot{r}$ and $r\dot{f}$ given in Eqs. (2.31) and (2.32). This gives

$$\begin{aligned}\dot{x} &= -\frac{na}{\sqrt{1-e^2}} \sin f, \\ \dot{y} &= +\frac{na}{\sqrt{1-e^2}} (e + \cos f).\end{aligned} \tag{2.36}$$

By comparing Eq. (2.34) with Eq. (2.28) we see that the energy constant can be written as

$$C = -\frac{\mu}{2a}, \tag{2.37}$$

and hence the energy of an elliptical orbit is a function of its semi-major axis alone and is independent of the eccentricity. Similar quantities can be defined for parabolic and hyperbolic orbits. It can be shown that

$$C_\mathrm{para} = 0, \quad \text{and} \quad C_\mathrm{hyper} = \frac{\mu}{2a}. \tag{2.38}$$

2.4 The Mean and Eccentric Anomalies

In the previous section we showed that, given the value of the true anomaly f, we can calculate the orbital radius and velocity of a body provided we know the eccentricity and semi-major axis of its orbit. However, in practice we usually want to calculate the location of a body at a given time and our solution to the two-body problem (Eq. (2.20)) does not contain the time explicitly. Although f and r are functions of t, we have not shown the nature of this dependence, although it is obviously nonlinear for $e \neq 0$.

Ideally we would like to make use of an angle that is not only 2π-periodic but also a linear function of the time. This will be particularly useful later on when we have to calculate time averages of various quantities. Using our definition of the mean motion n in Eq. (2.25) we can define the *mean anomaly* M by

$$M = n(t - \tau), \tag{2.39}$$

where the constant τ is the *time of pericentre passage*. Although M has the dimensions of an angle, and it increases linearly with time at a constant rate equal to the mean motion, it has no simple geometrical interpretation. However, from our definition of M and Eq. (2.20) it is clear that when $t = \tau$ (pericentre passage), $M = f = 0$, and when $t = \tau + T/2$ (apocentre passage), $M = f = \pi$; similar relationships will hold for additive multiples of the orbital period T.

Although M has no simple geometrical interpretation, it can be related to an angle that does. Consider a circumscribed circle, radius a, that is concentric with an orbital ellipse of semi-major axis a and eccentricity e (see Fig. 2.7). A line perpendicular to the major axis of the ellipse is extended through the point on the orbit and intersects the circle. We can define E, the *eccentric anomaly*, to be the angle between the major axis of the ellipse and the radius from the centre

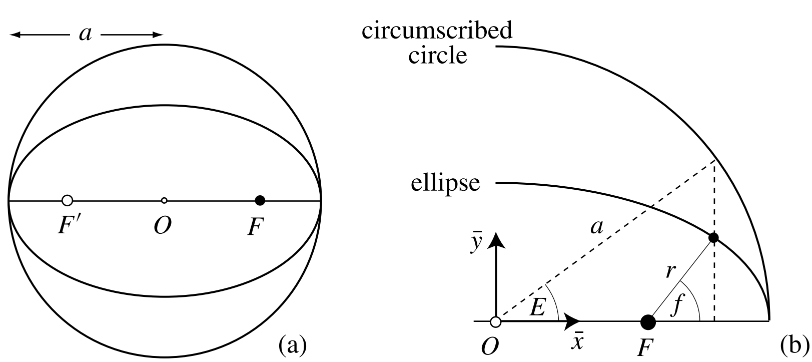

Fig. 2.7. (a) The circumscribed, concentric circle has a radius a equal to the semi-major axis of the ellipse. (b) The relationship between the true anomaly f and the eccentric anomaly E.

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to the intersection point on the circumscribed circle. Hence, $E = 0$ corresponds to $f = 0$ and $E = \pi$ corresponds to $f = \pi$.

The equation of a centred ellipse in rectangular coordinates is

$$\left(\frac{\bar{x}}{a}\right)^2 + \left(\frac{\bar{y}}{b}\right)^2 = 1. \tag{2.40}$$

But from Fig. 2.7 we have $\bar{x} = a \cos E$ and therefore $\bar{y}^2 = b^2 \sin^2 E$ and hence from Eq. (2.18), $\bar{y} = a\sqrt{1-e^2} \sin E$. Thus the projections of r in the horizontal and vertical directions are

$$x = a(\cos E - e) \qquad \text{and} \qquad y = a\sqrt{1-e^2} \sin E \tag{2.41}$$

(cf. Eq. (2.21)). By adding the squares of these expressions and then taking the square root we have

$$r = a(1 - e \cos E) \tag{2.42}$$

and

$$\cos f = \frac{\cos E - e}{1 - e \cos E}. \tag{2.43}$$

We can derive a simpler relationship between E and f by writing

$$1 - \cos f = \frac{(1+e)(1-\cos E)}{1 - e\cos E}, \qquad 1 + \cos f = \frac{(1-e)(1+\cos E)}{1 - e\cos E}. \tag{2.44}$$

Using the standard double angle formulae, these equations can be written as

$$2\sin^2\frac{f}{2} = \frac{1+e}{1-e\cos E} 2\sin^2\frac{E}{2}, \qquad 2\cos^2\frac{f}{2} = \frac{1-e}{1-e\cos E} 2\cos^2\frac{E}{2} \tag{2.45}$$

and hence

$$\tan\frac{f}{2} = \sqrt{\frac{1+e}{1-e}} \tan\frac{E}{2}. \tag{2.46}$$

Thus, knowing E, we can determine r and f uniquely from Eqs. (2.42) and (2.43), since E and f will always lie in the same half of the ellipse. However, to locate a body in its orbit at some time t, we need to derive a relationship between M and E.

Using $v^2 = \dot{r}^2 + (r\dot{f})^2$ and Eqs. (2.32) and (2.34), we have

$$\dot{r}^2 = n^2 a^3 \left(\frac{2}{r} - \frac{1}{a}\right) - \frac{n^2 a^4 (1-e^2)}{r^2}. \tag{2.47}$$

Hence

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{na}{r}\sqrt{a^2e^2 - (r-a)^2}. \tag{2.48}$$

This can then be integrated by making the substitution

$$r - a = -ae\cos E \tag{2.49}$$

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from Eq. (2.42). Hence Eq. (2.48) can be written as

$$\frac{\mathrm{d}E}{\mathrm{d}t} = \frac{n}{1 - e\cos E}\,. \tag{2.50}$$

This equation can also be derived by differentiating Eq. (2.41) to find $\dot{x}$ and $\dot{y}$, finding $\mathbf{r} \times \dot{\mathbf{r}}$ and equating the magnitude, h, with $na^2\sqrt{1-e^2}$. The resulting equation, Eq. (2.50), can be easily integrated to give

$$n(t - \tau) = E - e\sin E \tag{2.51}$$

where we have taken τ to be the constant of integration and used the boundary condition $E = 0$ when $t = \tau$. Hence, from Eq. (2.39) we have

$$M = E - e\sin E\,. \tag{2.52}$$

This is *Kepler's equation* and its solution is fundamental to the problem of finding the orbital position at a given time. At a particular time t we can (i) find M from Eq. (2.39), (ii) solve Eq. (2.52) for E, (iii) use Eq. (2.41), or Eqs. (2.43) and (2.20), to find r and f.

So far we have defined the true longitude (θ), the true anomaly (f), the mean anomaly (M), the eccentric anomaly (E), and the longitude of pericentre (ϖ). To complete this set we define the *mean longitude* λ by

$$\lambda = M + \varpi\,. \tag{2.53}$$

Therefore λ is a linear function of time and, since it is derived from M, it has no geometrical interpretation, except in the special case of a circular orbit. It is important to note that all longitudes (θ, ϖ, λ) are defined with respect to a common, arbitrary reference direction (see Fig. 2.5).

Colwell (1993) points out that papers have been published about the solution of Kepler's equation in virtually every decade since 1650 and that many eminent scientists have attempted solutions. Kepler's equation cannot be solved directly because it is transcendental in E and therefore, apart from the trivial solutions $E = j\pi$ when $M = j\pi$ for integer j, it is impossible to express E as a simple function of M. We briefly consider two iterative techniques: one producing a series solution and the other a numerical solution. In each case we assume that M and E are expressed in radians.

We can derive a series solution by using an iterative method of the form

$$E_{i+1} = M + e\sin E_i, \qquad i = 0, 1, \ldots, \tag{2.54}$$

where we take $E_0 = M$ as our first approximation. Using the formula $\sin(A + B) = \sin A\cos B + \cos A\sin B$ and the fact that $\sin x \approx x - \frac{1}{6}x^3 + \mathcal{O}(x^5)$ and

$\cos x \approx 1 - \frac{1}{2}x^2 + \mathcal{O}(x^4)$ for small x, we obtain

$$\begin{aligned}
E_1 &= M + e\sin M, \\
E_2 &= M + e\sin(M + e\sin M) \approx M + e\sin M + \frac{1}{2}e^2\sin 2M, \\
E_3 &= M + e\sin(M + e\sin M + \frac{1}{2}e^2\sin 2M) \\
&\approx M + \left(e - \frac{1}{8}e^3\right)\sin M + \frac{1}{2}e^2\sin 2M + \frac{3}{8}e^3\sin 3M
\end{aligned} \tag{2.55}$$

for the first three steps, where we introduced only one additional term in e at each step. It is clear from this approach that the final series for $E - M$ will have the form

$$E - M = \sum_{s=1}^{\infty} b_s(e)\sin sM, \tag{2.56}$$

where the lowest order term in $b_s(e)$ is $\mathcal{O}(e^s)$. The form of Eq. (2.56) suggests that we have expressed $E - M$ as a Fourier sine series in M. We study such useful expansions for elliptical motion in more detail in Sect. 2.5, where we derive expressions for the $b_s(e)$ terms.

It is important to note that the series solution of Kepler's equation diverges for values of $e > 0.6627434$ (see Hagihara 1970, for a detailed explanation). Not only is this property a fundamental limitation to deriving a useful series solution to Kepler's equation, it also has important implications for other series, such as the planetary disturbing function (see Chap. 6), that make use of this solution. However, numerical solutions of Kepler's equation are not affected by this limitation.

Danby (1988) gives a variety of numerical methods for solving Kepler's equation. By writing Eq. (2.52) as

$$f(E) = E - e\sin E - M \tag{2.57}$$

we can use the Newton–Raphson method to find the root of the nonlinear equation $f(E) = 0$. The iteration scheme is

$$E_{i+1} = E_i - \frac{f(E_i)}{f'(E_i)}, \qquad i = 0, 1, 2, \ldots, \tag{2.58}$$

where $f'(E_i) = \mathrm{d}f(E_i)/\mathrm{d}E_i = 1 - e\cos E_i$. Danby (1988) points out that the convergence of the Newton–Raphson scheme is quadratic but that quartic convergence is also possible with a modified scheme. Using Danby's notation and a Taylor series expansion we can write

$$0 = f(E_i + \epsilon_i) = f(E_i) + \epsilon_i f'(E_i) + \frac{1}{2}\epsilon_i^2 f''(E_i) + \frac{1}{6}\epsilon_i^3 f'''(E_i) + \mathcal{O}(\epsilon_i^4). \tag{2.59}$$

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Neglecting the higher order terms in ϵ_i we can write

$$0 = f_i + \delta_i f_i' + \frac{1}{2}\delta_i^2 f_i'' + \frac{1}{6}\delta_i^3 f_i''', \tag{2.60}$$

where $f_i = f(E_i)$, $f_i' = f'(E_i)$, etc. Hence

$$\delta_i = -\frac{f_i}{f_i' + \frac{1}{2}\delta_i f_i'' + \frac{1}{6}\delta_i^2 f_i'''}. \tag{2.61}$$

This can be solved for δ_i by defining

$$\delta_{i1} = -\frac{f_i}{f_i'}, \quad \delta_{i2} = -\frac{f_i}{f_i' + \frac{1}{2}\delta_{i1} f_i''}, \quad \delta_{i3} = -\frac{f_i}{f_i' + \frac{1}{2}\delta_{i2} f_i'' + \frac{1}{6}\delta_{i2}^2 f_i'''} \tag{2.62}$$

and then using the iteration scheme

$$E_{i+1} = E_i + \delta_{i3}. \tag{2.63}$$

Although this method has more arithmetic operations per iteration than the standard Newton–Raphson scheme given above, it is more efficient since (a) it can be programmed to make use of quantities that have already been calculated at each iteration and (b) it will converge faster.

An important consideration in either of these numerical schemes is a suitable starting value, E_0. Obviously for small e we have $E \approx M$ and so $E_0 = M$ seems appropriate. However, this guess is only correct in the cases where $e = 0$ or M is a multiple of π. Danby (1988) points out that, by first reducing M to the range $0 \le M \le 2\pi$, the initial guess

$$E_0 = M + \mathrm{sign}(\sin M)\, ke, \qquad 0 \le k \le 1 \tag{2.64}$$

has a better chance of being correct and improves the convergence; the recommended value is $k = 0.85$. Further details of this and other methods are discussed in Danby & Burkardt (1983), Burkardt & Danby (1983), and Danby (1987).

The solution of Kepler's equation to find E for a given value of M allows the calculation of the position and velocity at any time t for an object in an elliptical orbit. If the object has a position vector $\mathbf{r}_0 = \mathbf{r}(t_0)$ and a velocity vector $\mathbf{v}_0 = \mathbf{v}(t_0)$ at time t_0 then this process can be simplified by the introduction of two special functions and their time derivatives. Provided that the initial vectors $\mathbf{r}_0$ and $\mathbf{v}_0$ are not parallel, $\mathbf{r}(t)$ can be written as

$$\mathbf{r}(t) = f(t, t_0)\mathbf{r}_0 + g(t, t_0)\mathbf{v}_0, \tag{2.65}$$

where $f(t, t_0)$ and $g(t, t_0)$ are referred to as the *f and g functions.*

Separating the x and y components we have

$$x = f(t, t_0)x_0 + g(t, t_0)\dot{x}_0 \qquad \text{and} \qquad y = f(t, t_0)y_0 + g(t, t_0)\dot{y}_0, \tag{2.66}$$

where we have taken $\mathbf{r}_0 = (x_0, y_0)$ and $\mathbf{v}_0 = (\dot{x}_0, \dot{y}_0)$. This gives two simultaneous linear equations, which we can solve for f and g. The solution is

$$f(t, t_0) = \frac{x\dot{y}_0 - y\dot{x}_0}{x_0\dot{y}_0 - y_0\dot{x}_0} \qquad \text{and} \qquad g(t, t_0) = \frac{yx_0 - xy_0}{x_0\dot{y}_0 - y_0\dot{x}_0}. \tag{2.67}$$

Since $\cos f = x/r$ and $\sin f = y/r$ we can write Eq. (2.36) in terms of the eccentric anomaly instead of the true anomaly. This gives

$$\dot{x} = -\frac{na^2}{r}\sin E \qquad \text{and} \qquad \dot{y} = \frac{na^2\sqrt{1-e^2}}{r}\cos E\,. \tag{2.68}$$

Substituting Eqs. (2.41) and (2.68), with appropriate expressions for x_0, y_0, $\dot{x}_0$, and $\dot{y}_0$, and making use of Eqs. (2.42) and (2.51) gives

$$\begin{aligned} f(t, t_0) &= \frac{a}{r_0}\left\{\cos(E - E_0) - 1\right\} + 1, \\ g(t, t_0) &= (t - t_0) + \frac{1}{n}\left\{\sin(E - E_0) - (E - E_0)\right\}. \end{aligned} \tag{2.69}$$

The velocity at time t can be written as

$$\mathbf{v}(t) = \dot{f}\mathbf{r}_0 + \dot{g}\mathbf{v}_0, \tag{2.70}$$

where $\dot{f}$ and $\dot{g}$, the partial derivatives of f and g with respect to time, can be derived from Eq. (2.67) by making use of the expression for $\dot{E}$ given in Eq. (2.50). We then have

$$\begin{aligned} \dot{f}(t, t_0) &= -\frac{a^2}{rr_0}n\sin(E - E_0), \\ \dot{g}(t, t_0) &= \frac{a}{r}\left\{\cos(E - E_0) - 1\right\} + 1. \end{aligned} \tag{2.71}$$

The use of the f and g functions means that once E is known from the solution of Kepler's equation, we can readily find $\mathbf{r}$ and $\mathbf{v}$. Although we have formulated the expressions for the scalar quantities f, g, $\dot{f}$, and $\dot{g}$ by considering motion in the plane of the orbit, the formulae are equally applicable in other reference systems. In particular the f and g functions obviate the need to transform to and from a coordinate system in the orbital plane to one in a more general three-dimensional reference frame (see Sect. 2.8). This introduces considerable computational savings in numerical work.

2.5 Elliptic Expansions

Since there are so few integrable problems in solar system dynamics, frequently we have to resort to approximations in order to achieve a practical solution to a particular problem. The small quantities inherent in most branches of solar system dynamics are the eccentricity and inclination (the angle the orbit plane makes with a reference plane) of an orbit. For example, in Sect. 2.4 we have shown how Kepler's equation can be solved by means of a series in powers of the eccentricity. In Chapter 6 we deal with an expansion of the perturbing potential experienced by one planet or satellite due to another. In that case the expansion is in terms of the eccentricity and inclination of the bodies involved. Throughout this book we make use of a number of expansions. Typically we make an expansion, neglect higher order terms in some quantity, and then apply the resulting series to a problem of interest. In this section we derive a number of fundamental expansions that will be needed later on.

In the previous section we saw how it was possible to derive a simple series solution for E in terms of M. We can now formalise the result we obtained. If we write Eq. (2.52) as $E - M = e \sin E$ then, since $E - M$ is an odd periodic function, it can be expanded as a Fourier sine series,

$$e \sin E = \sum_{s=1}^{\infty} b_s(e) \sin sM, \tag{2.72}$$

where the coefficients $b_s(e)$ are given by

$$\begin{aligned} b_s(e) &= \frac{2}{\pi} \int_0^{\pi} e \sin E \sin sM \, \mathrm{d}M \\ &= \left[-\frac{2}{s\pi} e \sin E \cos sM \right]_0^{\pi} + \frac{2}{s\pi} \int_0^{\pi} \cos sM \, \mathrm{d}(e \sin E). \end{aligned} \tag{2.73}$$

The first part of this equation evaluates to zero and by using Kepler's equation to write $\mathrm{d}(e \sin E) = \mathrm{d}(E - M)$ we have

$$b_s(e) = -\frac{2}{s\pi} \int_0^{\pi} \cos sM \, \mathrm{d}M + \frac{2}{s\pi} \int_0^{\pi} \cos sM \, \mathrm{d}E. \tag{2.74}$$

The first integral evaluates to zero and we can use Kepler's equation again to obtain

$$b_s(e) = \frac{2}{s\pi} \int_0^{\pi} \cos(sE - se \sin E) \, \mathrm{d}E. \tag{2.75}$$

This integral can be written in terms of a standard function called the *Bessel function* of the first kind (see, e.g., Bowman 1958). We can write

$$b_s(e) = \frac{2}{s} J_s(se), \tag{2.76}$$

where

$$J_s(se) = \frac{1}{\pi} \int_0^{\pi} \cos(sE - se \sin E) \, \mathrm{d}E \tag{2.77}$$

is the Bessel function. For positive values of s, we can write

$$J_s(x) = \frac{1}{s!} \left(\frac{x}{2}\right)^s \sum_{\beta=0}^{\infty} (-1)^{\beta} \frac{(x/2)^{2\beta}}{\beta!(s+1)(s+2)\ldots(s+\beta)}. \tag{2.78}$$

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This series is absolutely convergent for all values of x. The series for $J_s(x)$ for $s = 1, \ldots, 5$ including terms up to $\mathcal{O}(x^5)$ are given below:

$$\begin{aligned}
J_1(x) &= \frac{1}{2}x - \frac{1}{16}x^3 + \frac{1}{384}x^5 + \mathcal{O}(x^7), \\
J_2(x) &= \frac{1}{8}x^2 - \frac{1}{96}x^4 + \mathcal{O}(x^6), \\
J_3(x) &= \frac{1}{48}x^3 - \frac{1}{768}x^5 + \mathcal{O}(x^7), \\
J_4(x) &= \frac{1}{384}x^4 + \mathcal{O}(x^6), \\
J_5(x) &= \frac{1}{3840}x^5 + \mathcal{O}(x^7).
\end{aligned} \tag{2.79}$$

We can now write the solution of Kepler's equation as

$$\begin{aligned}
E &= M + 2\sum_{s=1}^{\infty} \frac{1}{s} J_s(se) \sin sM \\
&= M + e\sin M + e^2\left(\frac{1}{2}\sin 2M\right) + e^3\left(\frac{3}{8}\sin 3M - \frac{1}{8}\sin M\right) \\
&\quad + e^4\left(\frac{1}{3}\sin 4M - \frac{1}{6}\sin 2M\right) + \mathcal{O}(e^5),
\end{aligned} \tag{2.80}$$

which is consistent with our result in Sect. 2.4. It is important to repeat the warning, mentioned in Sect. 2.4, that although this series solution rapidly converges for small values of e, the series is divergent for values of $e > 0.6627434$. This implies that all the series in this section that make use of this series are also divergent for sufficiently large values of e.

In addition to the series solution of Kepler's equation we will also need a number of other series expansions, all of which can be expressed in terms of Bessel functions. In particular we need series for r/a, $\cos E$, $(a/r)^3$, $\sin f$, $\cos f$, and $f - M$. The derivation of the results that follow are given in Brouwer & Clemence (1961).

The series for r/a is given by

$$\begin{aligned}
\frac{r}{a} &= 1 + \frac{1}{2}e^2 - 2e\sum_{s=1}^{\infty} \frac{1}{s^2}\frac{\mathrm{d}}{\mathrm{d}e} J_s(se)\cos sM \\
&= 1 - e\cos M + \frac{e^2}{2}(1 - \cos 2M) + \frac{3e^3}{8}(\cos M - \cos 3M) \\
&\quad + \frac{e^4}{3}(\cos 2M - \cos 4M) + \mathcal{O}(e^5).
\end{aligned} \tag{2.81}$$

This series is used in Sect. 2.6 in the guiding centre approximation and in Sect. 6.3 and 6.5 in our expansion of the planetary disturbing function.

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We can use the fact that $\cos E = (1 - r/a)/e$ and the series for (r/a) to derive the series for $\cos E$. It is given by

$$\begin{aligned}\cos E &= -\frac{1}{2}e + 2\sum_{s=1}^{\infty}\frac{1}{s^2}\frac{\mathrm{d}}{\mathrm{d}e}J_s(se)\cos sM \\ &= \cos M + \frac{e}{2}(\cos 2M - 1) + \frac{3e^2}{8}(\cos 3M - \cos M) \\ &\quad + e^3\left(\frac{1}{3}\cos 4M - \frac{1}{3}\cos 2M\right) \\ &\quad + e^4\left(\frac{5}{192}\cos M - \frac{45}{128}\cos 3M + \frac{125}{384}\cos 5M\right) + \mathcal{O}(e^5). \end{aligned} \tag{2.82}$$

We can also use the series for r/a to derive the series for $(a/r)^3$. It is given by

$$\begin{aligned}\left(\frac{a}{r}\right)^3 &= 1 + 3e\cos M + e^2\left(\frac{3}{2} + \frac{9}{2}\cos 2M\right) \\ &\quad + e^3\left(\frac{27}{8}\cos M + \frac{53}{8}\cos 3M\right) \\ &\quad + e^4\left(\frac{15}{8} + \frac{7}{2}\cos 2M + \frac{77}{8}\cos 4M\right) + \mathcal{O}(e^5). \end{aligned} \tag{2.83}$$

This series is used in Sect. 6.3 in our expansion of the planetary disturbing function.

The series for $\sin f$ and $\cos f$ are given by

$$\begin{aligned}\sin f &= 2\sqrt{1-e^2}\sum_{s=1}^{\infty}\frac{1}{s}\frac{\mathrm{d}}{\mathrm{d}e}J_s(se)\sin sM \\ &= \sin M + e\sin 2M + e^2\left(\frac{9}{8}\sin 3M - \frac{7}{8}\sin M\right) \\ &\quad + e^3\left(\frac{4}{3}\sin 4M - \frac{7}{6}\sin 2M\right) \\ &\quad + e^4\left(\frac{17}{192}\sin M - \frac{207}{128}\sin 3M + \frac{625}{384}\sin 5M\right) + \mathcal{O}(e^5) \end{aligned} \tag{2.84}$$

and

$$\begin{aligned}\cos f &= -e + \frac{2(1-e^2)}{e}\sum_{s=1}^{\infty}J_s(se)\cos sM \\ &= \cos M + e(\cos 2M - 1) + \frac{9e^2}{8}(\cos 3M - \cos M) \\ &\quad + \frac{4e^3}{3}(\cos 4M - \cos 2M) \\ &\quad + e^4\left(\frac{25}{192}\cos M - \frac{225}{128}\cos 3M + \frac{625}{384}\cos 5M\right) + \mathcal{O}(e^5). \end{aligned} \tag{2.85}$$

These series are used in Sect. 5.4 in our study of spin–orbit resonance and in Sect. 6.5 as part of the expansion of the planetary disturbing function.

We can derive a series for $f - M$, also called the *equation of the centre.* From Eqs. (2.20) and (2.32) we obtain

$$r^2 \dot{f} = na^2(1 - e^2)^{1/2}. \tag{2.86}$$

Using $\mathrm{d}M = n\,\mathrm{d}t$, $r = a(1 - e\cos E)$, and Kepler's equation, we obtain

$$\mathrm{d}f = \frac{\sqrt{1-e^2}}{(1 - e\cos E)^2}\mathrm{d}M = \sqrt{1-e^2}\left(\frac{\mathrm{d}E}{\mathrm{d}M}\right)^2 \mathrm{d}M. \tag{2.87}$$

This can be integrated term by term using the series solution for Kepler's equation to give

$$\begin{aligned} f - M = {} & 2e\sin M + \frac{5}{4}e^2\sin 2M + e^3\left(\frac{13}{12}\sin 3M - \frac{1}{4}\sin M\right) \\ & + e^4\left(\frac{103}{96}\sin 4M - \frac{11}{24}\sin 2M\right) + \mathcal{O}(e^5). \end{aligned} \tag{2.88}$$

This series is used in Sect. 2.6 in an analysis of the guiding centre approximation and in Sect. 5.2 where we investigate tidal de-spinning.

Lagrange developed a useful method for inverting series expansions, which has applications to this work. He showed that if a variable z can be expressed as a function of ζ of the form

$$\zeta = z + e\phi(\zeta) \qquad (e < 1) \tag{2.89}$$

then ζ can also be expressed as a function of z using the relationship

$$\zeta = z + \sum_{j=1}^{\infty} \frac{e^j}{j!}\frac{\mathrm{d}^{j-1}}{\mathrm{d}z^{j-1}}\left[\phi(z)\right]^j. \tag{2.90}$$

For example, to obtain the expression for f in terms of M, given in Eq. (2.88), we start from Kepler's second law, Eq. (2.8), and Eq. (2.26) giving,

$$h = r^2\dot{f} = na^2(1 - e^2)^{1/2}. \tag{2.91}$$

Integrating this equation and substituting Eq. (2.20) for r gives

$$M = (1 - e^2)^{3/2}\int_0^f \frac{\mathrm{d}f}{(1 + e\cos f)^2}. \tag{2.92}$$

Expanding binomially and integrating term by term, we obtain

$$M = f - 2e\sin f + \frac{3}{4}e^2\sin 2f + \mathcal{O}(e^3). \tag{2.93}$$

This can be written as

$$f = M + e\left(2\sin f - \frac{3}{4}e\sin 2f + \cdots\right). \tag{2.94}$$

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Lagrange's inverse theorem then gives

$$f = M + \sum_{j=1}^{\infty} \frac{e^j}{j!} \frac{\mathrm{d}^{j-1}}{\mathrm{d}M^{j-1}} \left[2 \sin M - \frac{3}{4} e \sin 2M + \cdots \right]^j , \tag{2.95}$$

which agrees with Eq. (2.88) after expansion. We make frequent use of Lagrange's method in Sect. 3.6 where we derive the locations of the collinear equilibrium points in the circular restricted problem.

2.6 The Guiding Centre Approximation

In many applications in solar system dynamics the eccentricity is very small and approximations that are accurate to order e are useful, particularly in some systems that are best viewed in a rotating reference frame. This approach is also appropriate when considering systems such as perturbed motion in the vicinity of equilibrium points (Sect. 3.8), the effects of planetary oblateness on near-circular, near-equatorial orbits (Sect. 6.11), and its applications to planetary rings (Chapter 10). In all these cases it is useful to characterise the extent of the departure from circular motion.

In the *guiding centre approximation,* the motion of a particle P moving in an elliptical orbit about a focus F (see Fig. 2.8) is viewed in a reference frame that is centred on a point G, the guiding centre, that rotates about the focus in a circle of radius a equal to the particle's semi-major axis, with angular speed equal to the particle's mean motion n.

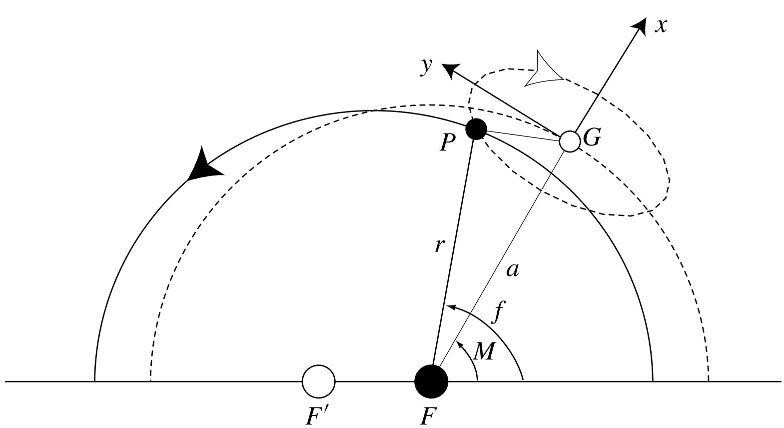

Fig. 2.8. The relationship between the true anomaly f and the mean anomaly M in the guiding centre approximation. G denotes the guiding centre, P the particle, F the focus, and F' the empty focus. The guiding centre moves on a circle of radius a centred on F.

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If we transform to a rectangular coordinate system centred on G, then the coordinates of P are

$$x = r\cos(f - M) - a \qquad \text{and} \qquad y = r\sin(f - M). \tag{2.96}$$

To order e, the expansion of $f - M$ from Eq. (2.88) is

$$f - M \approx 2e\sin M. \tag{2.97}$$

Hence

$$x \approx -ae\cos M \qquad \text{and} \qquad y \approx 2ae\sin M \tag{2.98}$$

and

$$\frac{x^2}{(ae)^2} + \frac{y^2}{(2ae)^2} \approx 1. \tag{2.99}$$

It follows that while G moves about F in a circle of radius a with mean motion n and period $2\pi/n$, P moves about G in the opposite sense on a 2:1 ellipse of semi-major axis $2ae$, semi-minor axis ae, and period $2\pi/n$. The motion of P with respect to F is a Lissajou figure obtained by the superposition of two harmonic motions with a common frequency n, a phase difference of $\pi/2$, and a 2:1 amplitude ratio.

At this level of approximation there are two other features of the motion that are worth noting. The distance R of P from the centre of the ellipse (see Fig. 2.9) can be obtained from

$$R^2 = r^2 + (ae)^2 + 2aer\cos f. \tag{2.100}$$

Hence

$$R \approx a\left(1 - \frac{1}{2}e^2\sin^2 f\right) \approx a\left(1 - \frac{1}{2}e^2\sin^2 M\right) \tag{2.101}$$

since, from Eq. (2.97), $f = M + \mathcal{O}(e)$. Thus, to order e, the path of P is a circle with centre at O. Therefore the path and the circumscribed circle (see Fig. 2.9) coincide and thus the angle $P\hat{O}F$ is the eccentric anomaly E. In fact, as we show below, in this approximation the angle $P\hat{F}'F$, where F' is the empty focus, is the mean anomaly M.

Consider the true elliptical path of P and denote the angle $F\hat{F}'P$ by g. Applying the cosine rule to the triangle $FF'P$ we obtain

$$r^2 = (2a - r)^2 + 4(ae)^2 - 4ae(2a - r)\cos g. \tag{2.102}$$

Hence

$$\cos g = \frac{(1 - r/a) + e^2}{e(1 - r/a) + e}. \tag{2.103}$$

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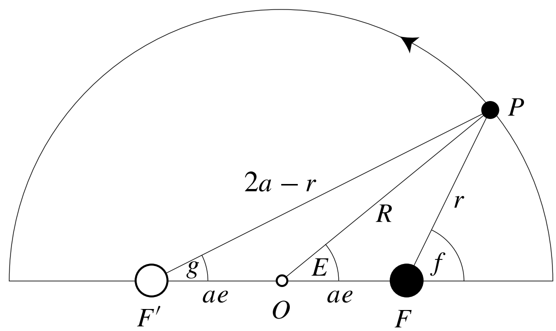

Fig. 2.9. The relationships among the true, mean, and eccentric anomalies in the guiding centre approximation. Note that the diagram exaggerates the actual case: In reality the eccentricity is small and F and F' are close to O.

Here we have used the fact that since F and F' are the foci of the ellipse, $FP + F'P = 2a$. From the expansion of r/a in Eq. (2.81) we obtain

$$1 - \frac{r}{a} \approx e \cos M - \frac{1}{2}e^2(1 - \cos 2M) - \frac{3}{8}e^3(\cos M - \cos 3M) \tag{2.104}$$

and hence

$$\cos g \approx \cos M - \frac{1}{8}e^2(\cos M - \cos 3M) + \mathcal{O}(e^3), \tag{2.105}$$

so that to $\mathcal{O}(e)$ we have $g = M$. Therefore the line joining the orbiting mass to the empty focus must rotate at the same rate as the mean motion of the orbiting mass.

This result has an interesting consequence if we apply it to the motion of a satellite that has a spin period equal to its orbital period (a *synchronously rotating* satellite). Since the line drawn from the satellite to the empty focus rotates with frequency n, equal to the mean motion, it follows that a synchronously rotating satellite rotates with one face pointing towards the empty focus of its orbit. This proves to be useful in understanding the origin of the librational tide on a synchronously rotating satellite like the Moon or Io. It also follows that the line drawn from the guiding centre to the central mass is parallel to the line joining the orbiting mass to the empty focus (Fig. 2.10).

It is interesting to note that Ptolemy's scheme for the motion of the Sun about the Earth had the Sun moving in a circle with uniform angular motion about an equant with the Earth displaced from the centre of the circle. If we place the Earth at the focus F and identify the equant with the point F', then we see that Ptolemy's scheme was accurate to order e. The triumph of Kepler was to produce a theory that was accurate to order e^2 (Hoyle 1974).

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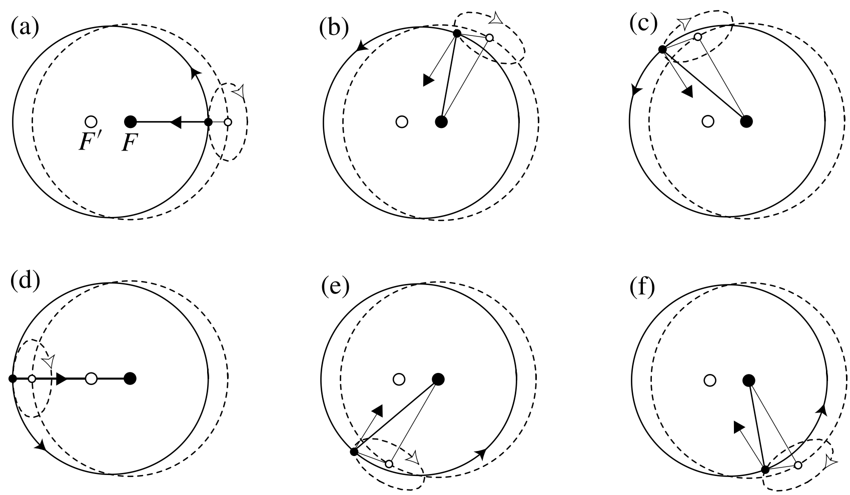

Fig. 2.10. An illustration of the guiding centre approximation for an ellipse of eccentricity $e = 0.2$. The position of the orbiting mass (small filled circle) with respect to the central mass (large filled circle) and the empty focus (large white circle) is shown at equal intervals of mean anomaly M. (a) $M = 0$, (b) $M = \pi/3$, (c) $M = 2\pi/3$, (d) $M = \pi$, (e) $M = 4\pi/3$, and (f) $M = 5\pi/3$. The solid curve denotes the keplerian ellipse while the dashed circle with a radius equal to the semi-major axis of the ellipse denotes the path of the guiding centre; the circle is centred on the primary focus of the ellipse.

2.7 Barycentric Orbits

We have shown that, with suitable starting conditions, the motion of the mass m_2 with respect to the mass m_1 describes a conic section in space. We now return to the formulation of the two-body problem using a centre of mass coordinate system with origin at the point O' (see Fig. 2.11).

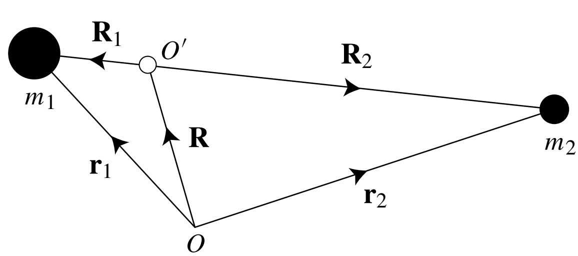

Fig. 2.11. The position vectors of the two masses with respect to the origin, O, and with respect to the centre of mass, O'.

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As before, let $\mathbf{R}$ denote the position vector of the centre of mass O', referred to the fixed origin O. The vector $\mathbf{R}$ is defined by the equation

$$m_1\mathbf{r}_1 + m_2\mathbf{r}_2 - (m_1 + m_2)\mathbf{R} = 0. \tag{2.106}$$

Because the sum of the coefficients in this equation is zero, the points defined by the three position vectors lie on a straight line. If we write

$$\mathbf{R}_1 = \mathbf{r}_1 - \mathbf{R} \qquad \text{and} \qquad \mathbf{R}_2 = \mathbf{r}_2 - \mathbf{R} \tag{2.107}$$

then it follows from the definition of $\mathbf{R}$ that

$$m_1\mathbf{R}_1 + m_2\mathbf{R}_2 = 0. \tag{2.108}$$

This implies that (i) $\mathbf{R}_1$ is always in the opposite direction to $\mathbf{R}_2$, and hence (ii) the centre of mass is always on the line joining m_1 and m_2, and we can write $R_1 + R_2 = r$, where r is the separation of m_1 and m_2, and (iii) the distances of the masses from the centre of mass are related by $m_1R_1 = m_2R_2$. It follows from this that

$$R_1 = \frac{m_2}{m_1 + m_2}r \qquad \text{and} \qquad R_2 = \frac{m_1}{m_1 + m_2}r. \tag{2.109}$$

Therefore, whichever conic section describes the relative motion of the two masses, each mass will also orbit the centre of mass of the system in a path described by the same conic section reduced in scale by $m_1/(m_1+m_2)$ or $m_2/(m_1+m_2)$ (see Fig. 2.12).

A coordinate system referred to the centre of mass is called a *barycentric* system. In Sect. 2.2 we showed that for the path of the relative motion of m_2

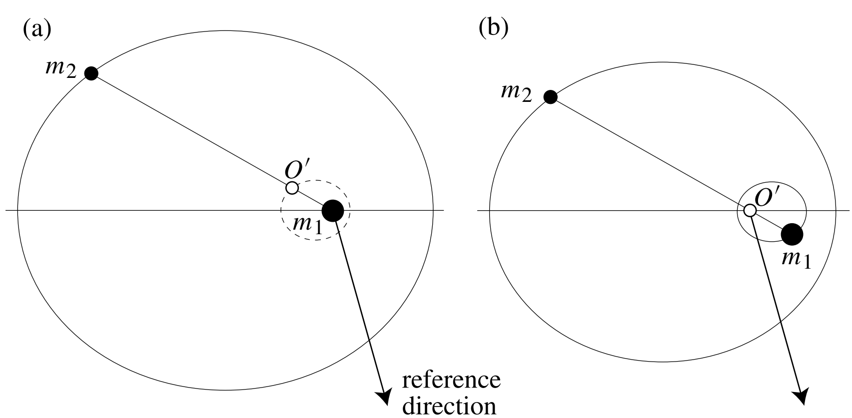

Fig. 2.12. (a) The motion of the mass m_2 with respect to the mass m_1 in the two-body problem; the dashed curve denotes the elliptical path of the centre of mass, O'. (b) The motion of the masses m_1 and m_2 with respect to the centre of mass, O'. In each case a mass ratio of $m_2/m_1 = 0.2$ and an eccentricity of 0.5 were used.

with respect to m_1, there exists a constant $h = r^2\dot{\theta}$. Since R_1 and R_2 are both proportional to r we have

$$
\begin{aligned}
R_1^2\dot{\theta} &= \text{constant} = h_1 = \left(\frac{m_2}{m_1+m_2}\right)^2 h, \\
R_2^2\dot{\theta} &= \text{constant} = h_2 = \left(\frac{m_1}{m_1+m_2}\right)^2 h,
\end{aligned}
\tag{2.110}
$$

and the total orbital angular momentum of the system is given by

$$
L^* = m_1 h_1 + m_2 h_2 = \frac{m_1 m_2}{m_1+m_2} h. \tag{2.111}
$$

Thus

$$
h = \left(\frac{1}{m_1} + \frac{1}{m_2}\right) L^* \tag{2.112}
$$

and hence, when $m_2 < m_1$, $h \approx h_2$ and h is approximately equal to the angular momentum of the system per unit mass of the body m_2.

It is clear from the above and Fig. 2.12 that the period of the orbit of each mass about the centre of mass is equal to T, the period of the mass m_2 about the mass m_1. Therefore the mean motions are also equal, $n_1 = n_2 = n$, although the semi-major axes are not. The relationship between the semi-major axes is suggested by considering Eq. (2.109) in the case of circular orbits; it is

$$
a_1 = \frac{m_2}{m_1+m_2} a \qquad \text{and} \qquad a_2 = \frac{m_1}{m_1+m_2} a. \tag{2.113}
$$

Consequently, since $h = na^2\sqrt{1-e^2}$, we have

$$
h_1 = na_1^2\sqrt{1-e^2} \qquad \text{and} \qquad h_2 = na_2^2\sqrt{1-e^2}, \tag{2.114}
$$

which shows that, although $m_1 a_1 = m_2 a_2$, the eccentricities of the ellipses are equal and therefore all the ellipses are similar. Although each mass moves on its own elliptical orbit with respect to the common centre of mass, the pericentres of their orbits differ by π (see Fig. 2.12b).

We can now consider the total energy of the system, E^*, which is the sum of the kinetic energy (referred to the inertial, barycentric coordinate system) and the potential energy. We have

$$
\begin{aligned}
E^* &= \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 - \mathcal{G}\frac{m_1 m_2}{r} \\
&= \frac{1}{2} m_1 \left[\dot{R}_1^2 + (R_1\dot{f})^2\right] + \frac{1}{2} m_2 \left[\dot{R}_2^2 + (R_2\dot{f})^2\right] - \mathcal{G}\frac{m_1 m_2}{r}.
\end{aligned}
\tag{2.115}
$$

We can simplify this using Eq. (2.109) to get

$$
E^* = \frac{m_1 m_2}{m_1+m_2} C = -\mathcal{G}\frac{m_1 m_2}{2a}, \tag{2.116}
$$

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where C is the energy constant from Eq. (2.37). Consequently, the total energy of the system is also purely a function of the semi-major axis of the orbit of m_2 with respect to m_1. Note that

$$C = \left(\frac{1}{m_1} + \frac{1}{m_2}\right) E^* \tag{2.117}$$

and hence, when $m_2 \ll m_1$, $C \approx E^*/m_2$, that is, the total energy per unit mass of the body m_2.

2.8 The Orbit in Space

In Sect. 2.2 we showed that the position and velocity vectors of the mass m_2 with respect to the mass m_1 always lie in a plane perpendicular to the angular momentum vector. The values of $\mathbf{r} = (x, y)$ and $\dot{\mathbf{r}} = (\dot{x}, \dot{y})$ (or alternatively, r, θ, $\dot{r}$, and $\dot{\theta}$) of the mass m_2 with respect to m_1 at any time define a unique orbit and a location on that orbit by means of the three constants a, e, and ϖ and the variable f. Our subsequent analysis was concerned with understanding the motion in the orbital plane. However, motion in the solar system is not confined to a single plane and we now consider the three-dimensional representation of an orbit in space (see Fig. 2.13).

Although we have shown that motion is confined to a fixed orbital plane, we consider a three-dimensional Cartesian coordinate system with respect to which an arbitrary point has a position vector $\mathbf{r} = (x, y, z) = x\hat{\mathbf{x}} + y\hat{\mathbf{y}} + z\hat{\mathbf{z}}$. The x axis is taken to lie along the major axis of the ellipse in the direction of pericentre, the y axis is perpendicular to the x axis and lies in the orbital plane, while the

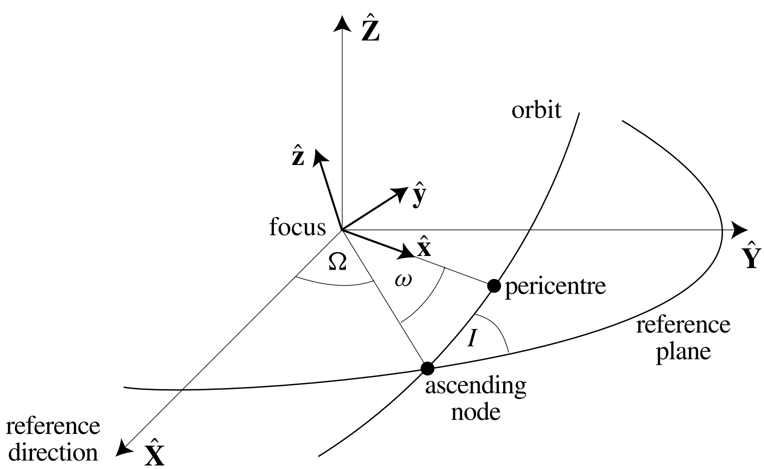

Fig. 2.13. Orbital motion with respect to the reference plane in three-dimensional space.

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z axis is mutually perpendicular to the x and y axes such that all three form a right-handed triad.

We now wish to refer this orbital plane to a standard reference plane. The direction of the reference line in the reference plane forms the X axis of our standard coordinate system. The Y axis is in the reference plane at right angles to the X axis, while the Z axis is perpendicular to both the X and Y axes forming a right-handed triad. For example, when considering the motion of planets around the Sun, it is customary to use a Sun-centred, or *heliocentric coordinate system* where the reference plane is the plane of the Earth's orbit (the *ecliptic*) and the reference line is in the direction of the *vernal equinox*, along the line of intersection of the plane of the Earth's equator and the ecliptic. It should be pointed out that this particular reference frame varies with time because of perturbations by other bodies (see, for example, Standish et al. (1992) for a thorough discussion of coordinate systems and reference frames or Montenbruck (1989) for a set of coordinate transformations).

In general the orbital plane will be inclined to the reference plane at an angle I called the *inclination* of the orbit. The line of intersection between the orbital plane and the standard reference frame is called the *line of nodes*. The point in both planes where the orbit crosses the reference plane moving from below to above the plane is called the *ascending node* while the angle between the reference line and the radius vector to the ascending node is called the *longitude of ascending node*, Ω. The angle between this same radius vector and the pericentre of the orbit is called the *argument of pericentre*, ω.

The inclination is always in the range $0 \leq I \leq 180^\circ$. If $I < 90^\circ$ the motion is said to be *prograde* whereas if $I \geq 90^\circ$ the motion is *retrograde*. In the limit as $I \rightarrow 0$ the orbital plane coincides with the reference plane and we have

$$\varpi = \Omega + \omega, \tag{2.118}$$

where ϖ is the longitude of pericentre, which was introduced in Sect. 2.3. However, the definition of ϖ in (2.118) is also used in the inclined case, despite the fact that the angles Ω and ω lie in different planes. In general, therefore, ϖ is a "dogleg" angle.

Figure 2.14 shows the relationship between the orbital plane coordinate system and the reference plane system. It is clear that coordinates in one system can be expressed in terms of the other by means of a series of three rotations about various axes.

To transform from the (x, y, z) orbital plane system to the general (X, Y, Z) reference system we have to carry out (i) a rotation about the z axis through an angle ω so that the x axis coincides with the line of nodes, (ii) a rotation about the x axis through an angle I so that the two planes are coincident, and finally (iii) a rotation about the z axis through an angle Ω (see Fig. 2.13 and Fig. 2.14).

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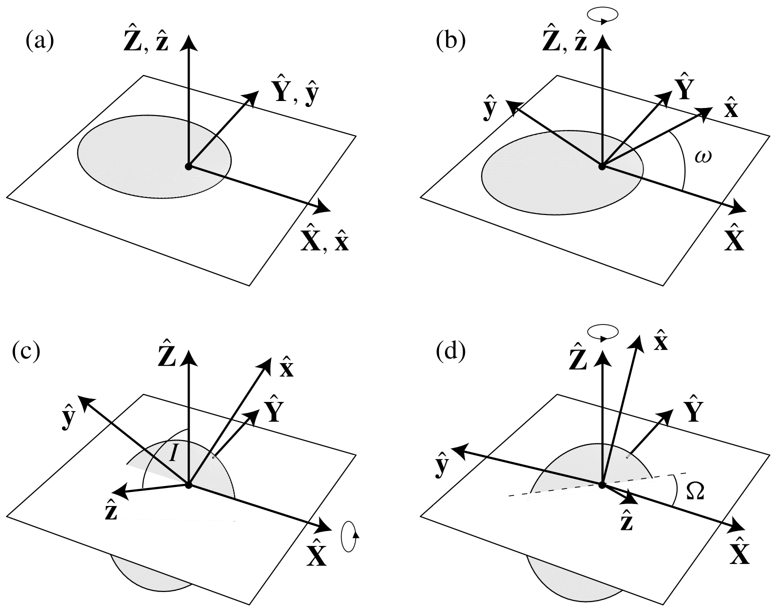

Fig. 2.14. The relationship between the unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$, $\hat{\mathbf{X}}$, $\hat{\mathbf{Y}}$, $\hat{\mathbf{Z}}$ and the angles ω, I, and Ω. (a) The transformation can be achieved through a series of three rotations applied to originally coincident axes. (b) The first rotation is through a positive angle ω about the $\hat{\mathbf{Z}}$ axis. (c) The second is through a positive angle I about the $\hat{\mathbf{X}}$ axis. (d) The final rotation is through a positive angle Ω about the $\hat{\mathbf{Z}}$ axis.

We can represent these transformations by three 3×3 rotation matrices, denoted by $\mathbf{P}_1$, $\mathbf{P}_2$, and $\mathbf{P}_3$ respectively with elements

$$\mathbf{P}_1 = \begin{pmatrix} \cos\omega & -\sin\omega & 0 \\ \sin\omega & \cos\omega & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \mathbf{P}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos I & -\sin I \\ 0 & \sin I & \cos I \end{pmatrix}, \tag{2.119}$$

and

$$\mathbf{P}_3 = \begin{pmatrix} \cos\Omega & -\sin\Omega & 0 \\ \sin\Omega & \cos\Omega & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{2.120}$$

Consequently,

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \mathbf{P}_3\mathbf{P}_2\mathbf{P}_1 \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \mathbf{P}_1^{-1}\mathbf{P}_2^{-1}\mathbf{P}_3^{-1} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}, \tag{2.121}$$

where $\mathbf{P}_1^{-1}$ is the inverse of the matrix $\mathbf{P}_1$ etc. Because all rotation matrices are orthogonal, the inverse of each matrix is simply equal to its transpose.

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If we now restrict ourselves to coordinates that lie in the orbital plane, we have

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \mathbf{P}_3\mathbf{P}_2\mathbf{P}_1 \begin{pmatrix} r\cos f \\ r\sin f \\ 0 \end{pmatrix}$$
$$= r\begin{pmatrix} \cos\Omega\cos(\omega+f) - \sin\Omega\sin(\omega+f)\cos I \\ \sin\Omega\cos(\omega+f) + \cos\Omega\sin(\omega+f)\cos I \\ \sin(\omega+f)\sin I \end{pmatrix}. \tag{2.122}$$

Note that the values of a and e are unchanged by considering the ellipse in this new coordinate system, since rotational transformations preserve lengths.

As an example of the use of these formulae we consider the problem of finding the positions of the planets at a given time, say September 25, 1993 at 5.32 PM British Summer Time. Appendix A gives formulae for the calculation of the orbital elements of the planets at any time referred to the mean ecliptic and equinox of the epoch of noon on 1st January 2000; this is called the *J2000 epoch*. The formulae give the corrections as a function of T, the interval in centuries between the given date, and the J2000 epoch date. In these calculations it is convenient to express each date as a *Julian date*; this is the number of days measured from noon on 1st January 4713 B.C. A Julian century is defined to have 36,525 days. The Julian date of the J2000 epoch is 2451545.0 and that of our given date is 2449256.189, so in our case $T = -0.06266423$.

If we consider the orbital elements for one planet, say Jupiter, the formulae in Appendix A give $a_{\rm j} = 5.20332$ AU, $e_{\rm j} = 0.0484007$, $I_{\rm j} = 1^\circ\!.30537$, $\Omega_{\rm j} = 100^\circ\!.535$, $\varpi_{\rm j} = 14^\circ\!.7392$, and $\lambda_{\rm j} = 204^\circ\!.234$, where the subscript j refers to the values for Jupiter. Hence $M_{\rm j} = \lambda_{\rm j} - \varpi_{\rm j} = 189^\circ\!.495$. The numerical solution of Kepler's equation (see Sect. 2.4) gives $E_{\rm j} = 189^\circ\!.059$ and hence, from Eq. (2.41), we have

$$x_{\rm j} = -5.39027\,\text{AU} \qquad \text{and} \qquad y_{\rm j} = -0.818277\,\text{AU}. \tag{2.123}$$

Substituting the values of $I_{\rm j}$, $\Omega_{\rm j}$, and $\varpi_{\rm j}$ in Eqs. (2.119) and (2.120) we obtain

$$\mathbf{P}_{\rm j} = \mathbf{P}_3\mathbf{P}_2\mathbf{P}_1 = \begin{pmatrix} 0.966839 & -0.254401 & 0.0223971 \\ 0.254373 & 0.967097 & 0.00416519 \\ -0.0227198 & 0.00167014 & 0.99974 \end{pmatrix} \tag{2.124}$$

for the transformation matrix, and hence the coordinates of Jupiter in the J2000 reference frame are

$$X_{\rm j} = -5.00336, \qquad Y_{\rm j} = -2.16249, \qquad Z_{\rm j} = 0.121099. \tag{2.125}$$

The procedure can be applied to find the positions of the other planets. The results are illustrated in Fig. 2.15.

We can now summarise an algorithm for transforming the position (X, Y, Z) and velocity $(\dot{X}, \dot{Y}, \dot{Z})$ of an object in an elliptical orbit in the standard reference

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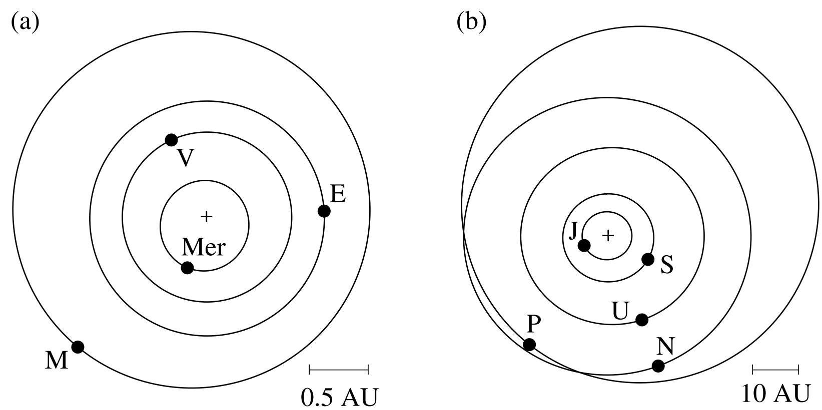

Fig. 2.15. The positions and orbits of the planets on September 25, 1993 at 5.32 PM BST, projected onto the ecliptic of J2000. (a) The inner solar system showing the positions of the planets Mercury (Mer), Venus (V), Earth (E), and Mars (M). (b) The outer solar system showing the positions of the planets Jupiter (J), Saturn (S), Uranus (U), Neptune (N), and Pluto (P). The filled circles denote the positions of the planets and are not to scale. The Sun is denoted by the cross. The lines show the scale of each plot.

plane at a time t to a set of six orbital elements, a, e, I, Ω, ω, and f, and a time of pericentre passage, τ. We assume that the masses of the central and orbiting objects are m_1 and m_2 respectively. We have

$$R^2 = X^2 + Y^2 + Z^2, \tag{2.126}$$
$$V^2 = \dot{X}^2 + \dot{Y}^2 + \dot{Z}^2, \tag{2.127}$$
$$\mathbf{R} \cdot \dot{\mathbf{R}} = X\dot{X} + Y\dot{Y} + Z\dot{Z}, \tag{2.128}$$
$$\mathbf{h} = (Y\dot{Z} - Z\dot{Y},\ Z\dot{X} - X\dot{Z},\ X\dot{Y} - Y\dot{X}), \tag{2.129}$$
$$\dot{R} = \pm\sqrt{V^2 - \frac{h^2}{R^2}}, \tag{2.130}$$

where $R = r$ now denotes the length of the radius vector and $\dot{R}$ is its rate of change. The sign of $\dot{R}$ is taken to be the sign of $\mathbf{R} \cdot \dot{\mathbf{R}}$, since R is always positive. Taking the projections of $\mathbf{h} = (h_X, h_Y, h_Z)$ onto the three planes we have

$$h \cos I = h_Z, \tag{2.131}$$
$$h \sin I \sin \Omega = \pm h_X, \tag{2.132}$$
$$h \sin I \cos \Omega = \mp h_Y, \tag{2.133}$$

where the upper sign in Eqs. (2.132) and (2.133) is taken if $h_Z > 0$ and the lower sign is taken if $h_Z < 0$.

The procedure is as follows:

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1) Calculate a using Eqs. (2.34), (2.126), and (2.127):

$$a = \left(\frac{2}{R} - \frac{V^2}{\mathcal{G}(m_1 + m_2)} \right)^{-1}. \tag{2.134}$$

2) Calculate e using Eqs. (2.26) and (2.134):

$$e = \sqrt{1 - \frac{h^2}{\mathcal{G}(m_1 + m_2)a}}. \tag{2.135}$$

3) Calculate I using Eq. (2.131):

$$I = \cos^{-1}\left(\frac{h_Z}{h} \right). \tag{2.136}$$

4) Calculate Ω using the expressions for $\sin\Omega$ and $\cos\Omega$ given in Eqs. (2.132) and (2.133):

$$\sin\Omega = \frac{\pm h_X}{h \sin I} \qquad \text{and} \qquad \cos\Omega = \frac{\mp h_Y}{h \sin I}. \tag{2.137}$$

The choice of sign is determined by the sign of h_Z (see above).

5) Calculate $\omega + f$ from the expressions for Z/R and X/R in Eq. (2.122), recalling that $r = R$:

$$\begin{aligned} \sin(\omega + f) &= \frac{Z}{R \sin I}, \\ \cos(\omega + f) &= \sec\Omega \left(\frac{X}{R} + \sin\Omega \sin(\omega + f) \cos I \right). \end{aligned} \tag{2.138}$$

6) Calculate f and hence ω from the expressions for $\sin f$ and $\cos f$ derived from Eqs. (2.20) and (2.31), recalling that $\dot{r} = \dot{R}$:

$$\sin f = \frac{a(1 - e^2)}{he} \dot{R} \qquad \text{and} \qquad \cos f = \frac{1}{e} \left(\frac{a(1 - e^2)}{R} - 1 \right). \tag{2.139}$$

7) Calculate τ by first calculating E from Eq. (2.42) and then using Eqs. (2.26) and (2.51):

$$\tau = t - \frac{E - e \sin E}{\sqrt{\mathcal{G}(m_1 + m_2)a^{-3}}}. \tag{2.140}$$

Although these equations define a correct procedure for deriving orbital elements, for numerical purposes it is desirable to eliminate $\mathcal{G}(m_1 + m_2)$ from the equations and choose a more practical system of units rather than the SI or any other standard system. This can be achieved by scaling the independent variable, t, by a factor $\sqrt{\mu} = \sqrt{\mathcal{G}(m_1 + m_2)}$ and using a new time variable, $\bar{t}$, such that

$$\sqrt{\mu}\, \mathrm{d}t = \mathrm{d}\bar{t}. \tag{2.141}$$

From Eq. (2.5) it can be seen that this has the same effect as setting $\mu = 1$. If, in addition, the unit of length is chosen so that $a = 1$ then we are dealing with a two-body system in which the orbit has unit mean motion and an orbital period of 2π time units. This is a common system of units to adopt when dealing with the circular restricted three-body problem (see Chapter 3).

2.9 Perturbed Orbits

We have seen in Sect. 2.8 that in the two-body problem the orbital elements a, e, I, ω, Ω, and τ are constants that are uniquely determined from the position and velocity of the orbiting mass. Even if there is a perturbing force acting on the system, any instantaneous set of position and velocity vectors always defines a set of six orbital elements that would give the shape and orientation of the orbit that the mass would follow if the perturbing force were to disappear at that instant. These are called *osculating elements*, from the Latin verb *osculare*, to kiss. Although we shall deal with perturbations to orbits in future sections, at this stage it is useful to examine some of the basic effects that perturbing forces exert on orbits.

Burns (1976) showed how the equations for the time derivatives of a, e, I, ω, Ω, and τ could be derived in a straightforward manner using elementary dynamics. Following his approach we consider a small disturbing force

$$\mathrm{d}\mathbf{F} = \bar{R}\hat{\mathbf{r}} + \bar{T}\hat{\boldsymbol{\theta}} + \bar{N}\hat{\mathbf{z}}, \tag{2.142}$$

where $\bar{R}$, $\bar{T}$, and $\bar{N}$ are the magnitudes of the radial, tangential, and normal components of the force respectively and $\hat{\mathbf{r}}$, $\hat{\boldsymbol{\theta}}$, and $\hat{\mathbf{z}}$ are the standard unit vectors introduced in Sect. 2.2. In the remainder of this section we derive expressions for $\dot{a}$, $\dot{e}$, $\dot{I}$, $\dot{\omega}$, $\dot{\Omega}$, and $\dot{\tau}$ as functions of these components and show which parts of the force give rise to changes in particular orbital elements.

We can equate the time derivative of the energy constant, C, with the work done on the orbiting body per unit mass per unit time. Hence

$$\dot{C} = \dot{\mathbf{r}} \cdot \mathrm{d}\mathbf{F} = \dot{r}\bar{R} + r\dot{\theta}\bar{T} \tag{2.143}$$

and, from Eq. (2.37),

$$\dot{C} = \frac{\mu}{2a^2}\dot{a}\,. \tag{2.144}$$

Hence, from the definitions of $\dot{r}$ and $r\dot{\theta}(= r\dot{f})$ in Eqs. (2.31) and (2.32),

$$\frac{\mathrm{d}a}{\mathrm{d}t} = 2\frac{a^{3/2}}{\sqrt{\mu(1-e^2)}}\left[\bar{R}e\sin f + \bar{T}(1+e\cos f)\right]. \tag{2.145}$$

This implies that only forces in the plane of the orbit can change its semi-major axis.

Using Eqs. (2.135) and (2.37) we can write

$$e = \sqrt{1 + 2h^2C\mu^{-2}} \tag{2.146}$$

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and hence

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \frac{e^2 - 1}{2e}(2\dot{h}/h + \dot{C}/C)\,. \tag{2.147}$$

Since the rate of change of angular momentum is equal to the applied moment, we have

$$\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t} = \mathbf{r} \times \mathrm{d}\mathbf{F} = r\bar{T}\hat{\mathbf{z}} - r\bar{N}\hat{\boldsymbol{\theta}}\,. \tag{2.148}$$

This implies that

$$\frac{\mathrm{d}h}{\mathrm{d}t} = r\bar{T} \tag{2.149}$$

since the $-r\bar{N}\hat{\boldsymbol{\theta}}$ component changes the direction of $\mathbf{h}$ but does not affect its magnitude. From Eq. (2.42) and the formulae for C, h, $\dot{C}$, $\dot{a}$, and $\dot{h}$ in Eqs. (2.37), (2.26), (2.144), (2.145), and (2.149) we have

$$\frac{\mathrm{d}e}{\mathrm{d}t} = \sqrt{a\mu^{-1}(1-e^2)}\left[\bar{R}\sin f + \bar{T}(\cos f + \cos E)\right]. \tag{2.150}$$

This implies that the eccentricity can only be changed by the application of forces in the orbital plane.

Differentiating Eq. (2.131) gives

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{\dot{h}/h - \dot{h}_Z/h_Z}{\sqrt{(h/h_Z)^2 - 1}}\,. \tag{2.151}$$

We can express the X, Y, and Z components of $\dot{\mathbf{h}}$ using

$$\begin{pmatrix} \dot{h}_X \\ \dot{h}_Y \\ \dot{h}_Z \end{pmatrix} = \mathbf{P}_3\mathbf{P}_2 \begin{pmatrix} \cos(\omega+f) & -\sin(\omega+f) & 0 \\ \sin(\omega+f) & \cos(\omega+f) & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ -r\bar{N} \\ r\bar{T} \end{pmatrix}, \tag{2.152}$$

where the matrices $\mathbf{P}_2$ and $\mathbf{P}_3$ are given in Eqs. (2.119) and (2.120). This gives

$$\begin{aligned} \dot{h}_X = r(&\bar{T}\sin I \sin\Omega + \bar{N}\sin(\omega+f)\cos\Omega \\ &+ \bar{N}\cos(\omega+f)\cos I \sin\Omega), \end{aligned} \tag{2.153}$$

$$\begin{aligned} \dot{h}_Y = r(&-\bar{T}\sin I \cos\Omega + \bar{N}\sin(\omega+f)\sin\Omega \\ &- \bar{N}\cos(\omega+f)\cos I \cos\Omega), \end{aligned} \tag{2.154}$$

$$\dot{h}_Z = r(\bar{T}\cos I - \bar{N}\cos(\omega+f)\sin I). \tag{2.155}$$

Substituting Eqs. (2.26), (2.131), (2.149), and (2.155) in Eq. (2.151) gives

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{\sqrt{a\mu^{-1}(1-e^2)}\bar{N}\cos(\omega+f)}{1 + e\cos f}, \tag{2.156}$$

which can also be written as

$$\frac{\mathrm{d}I}{\mathrm{d}t} = \frac{r\bar{N}\cos(\omega + f)}{h}\,. \tag{2.157}$$

Since $\mathrm{d}I/\mathrm{d}t$ only depends on $\bar{N}$, only forces normal to the orbital plane can change the inclination. Note that $r\bar{N}\cos(\omega + f)$ is the component of the torque that rotates the angular momentum vector about the line of nodes.

Dividing Eq. (2.132) by Eq. (2.133) gives

$$\tan\Omega = -h_X/h_Y, \tag{2.158}$$

which can then be differentiated with respect to time to give

$$\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{h_X\dot{h}_Y - h_Y\dot{h}_X}{h^2 - h_Z^2} \tag{2.159}$$

or

$$\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{\sin\Omega\dot{h}_Y + \cos\Omega\dot{h}_X}{h\sin I}\,. \tag{2.160}$$

Substituting Eqs. (2.26), (2.20), (2.153), and (2.154) into Eq. (2.160) gives

$$\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \sqrt{a\mu^{-1}(1 - e^2)}\frac{\bar{N}\sin(\omega + f)}{\sin I(1 + e\cos f)} \tag{2.161}$$

or

$$\frac{\mathrm{d}\Omega}{\mathrm{d}t} = \frac{r\bar{N}\sin(\omega + f)}{h\sin I}\,. \tag{2.162}$$

In this equation $r\bar{N}\sin(\omega + f)$ is the moment acting to precess the plane of the orbit and $h\sin I$ is the component of the angular momentum vector that lies normal to the line of nodes in the X–Y plane.

In order to derive an expression for $\dot{\omega}$ it is necessary to return to the equation of the ellipse, Eq. (2.20), and make use of our expressions for e and h given in Eqs. (2.146) and (2.26). This gives

$$h^2 = \mu r\left[1 + \sqrt{1 + 2Ch^2\mu^{-2}}\cos(\theta - \omega)\right], \tag{2.163}$$

where $\theta = \omega + f$ and we are choosing θ to be the position angle measured from the line of nodes. If we are interested in the change in the orbital elements due to the instantaneous application of a perturbing force $\mathrm{d}\mathbf{F}$ then C, $\mathbf{h}$, and ω change but r remains fixed. Differentiation of Eq. (2.163) gives

$$\begin{aligned}\frac{\mathrm{d}\omega}{\mathrm{d}t} = 2h\dot{h}\frac{r^{-1} + C(e\mu)^{-1}\cos(\theta - \omega)}{e\mu\sin(\theta - \omega)}\\ + \dot{\theta} - \frac{h^2}{e^2\mu^2}\dot{C}\cot(\theta - \omega)\,.\end{aligned} \tag{2.164}$$

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Substituting Eqs. (2.144) and (2.149) in Eq. (2.164) gives

$$\frac{\mathrm{d}\omega}{\mathrm{d}t} = e^{-1}\sqrt{a\mu^{-1}(1-e^2)}\left[-\bar{R}\cos f + \bar{T}\sin f\frac{2+e\cos f}{1+e\cos f}\right] - \dot{\Omega}\cos I \,. \tag{2.165}$$

The last term arises from the $\dot{\theta}$ term in Eq. (2.164) using the fact that the instantaneous change in θ is due to the change in the longitude of ascending node, since θ is referred to the nodal position (Burns 1976).

The equation for $\dot{\tau}$ is derived from the differentiation of Kepler's equation, Eq. (2.51). Setting $\chi = n\tau$ we have

$$\frac{\mathrm{d}\chi}{\mathrm{d}t} = \frac{\dot{C}}{C}\left(-\frac{3}{2}nt + \frac{(1-e^2)^{3/2}(2e-\cos f - e\cos^2 f)}{2e^2\sin f(1+e\cos f)}\right) - \frac{\dot{h}}{h}\frac{(1-e^2)^{3/2}}{e^2}\cot f \,. \tag{2.166}$$

Writing $\dot{\chi} = -n\dot{\tau} - \dot{n}\tau$ we have

$$\begin{aligned}\frac{\mathrm{d}\tau}{\mathrm{d}t} = &\left[3(\tau - t)\frac{\sqrt{a}}{\sqrt{\mu(1-e^2)}}e\sin f \right.\\ &\left. +a^2\mu^{-1}(1-e^2)\left(\frac{-\cos f}{e} + \frac{2}{1+e\cos f}\right)\right]\bar{R}\\ &+\left[3(\tau - t)\frac{\sqrt{a}}{\sqrt{\mu(1-e^2)}}(1+e\cos f)\right.\\ &\left. +a^2\mu^{-1}(1-e^2)\left(\frac{\sin f(2+e\cos f)}{e(1+e\cos f)}\right)\right]\bar{T} \,.\end{aligned} \tag{2.167}$$

Again only forces in the orbital plane can change the time of pericentre passage.

Note that the time t occurs on the right-hand side of Eqs. (2.166) and (2.167). This causes practical difficulties in the use of these derivatives since they grow with time. This problem also occurs with other forms of the perturbation equations and it will be discussed further in Sect. 6.8.

2.10 Hamiltonian Formulation

The approach taken in formulating and solving the equations of motion of the two-body problem and its perturbed motion is not unique and alternative formulations are not only possible but in some circumstances preferable. For most of the applications discussed in this book the classical approach is adequate, but there are some topics, notably the derivation of Lagrange's equations (Sect. 6.7), the discussion of the dynamics of resonance (Sect. 8.8), resonance

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encounters (Sect. 8.12), and algebraic mappings (Sect. 9.5), where a different mathematical approach is required. This is why we demonstrate the Hamiltonian formulation of the two-body problem in this section.

In Sect. 2.2 we formulated the equations of motion of the two-body problem (two objects of mass m_1 and m_2 moving under their mutual gravitational attraction) in terms of the Cartesian position (x, y) and velocity $(\dot{x}, \dot{y})$ of m_2 with respect to m_1, and we derived the differential equation

$$\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} + \mu\frac{\mathbf{r}}{r^3} = 0, \tag{2.168}$$

the solution of which is a conic section. Early on in our analysis we chose the reference plane to be the orbit plane. However, this vector equation is equally applicable to motion in three dimensions, with $\mathbf{r} = (x, y, z)$ and $\dot{\mathbf{r}} = (\dot{x}, \dot{y}, \dot{z})$. In what follows we use the variables $\mathbf{r}$ and $\mathbf{p}$ where, using a slightly different notation,

$$\mathbf{r} = r_x\mathbf{i} + r_y\mathbf{j} + r_z\mathbf{k} \qquad \text{and} \qquad \mathbf{p} = p_x\mathbf{i} + p_y\mathbf{j} + p_z\mathbf{k}. \tag{2.169}$$

Here $\mathbf{r}$ is the relative position vector, $\mathbf{p} = [m_1m_2/(m_1 + m_2)]\mathbf{v}$ is the linear momentum of the system, and, as usual, $\mathbf{v} = \dot{\mathbf{r}}$ is the velocity.

Now we can write the vector equations of motion in the form

$$\dot{\mathbf{r}} = +\nabla_{\mathbf{p}}\mathcal{H}_{\text{Kepler}} \qquad \text{and} \qquad \dot{\mathbf{p}} = -\nabla_{\mathbf{r}}\mathcal{H}_{\text{Kepler}} \tag{2.170}$$

where $\nabla_{\mathbf{p}}$ and $\nabla_{\mathbf{r}}$ are the vector differential operators given by

$$\nabla_{\mathbf{p}} = \mathbf{i}\frac{\partial}{\partial p_x} + \mathbf{j}\frac{\partial}{\partial p_y} + \mathbf{k}\frac{\partial}{\partial p_z} \quad \text{and} \quad \nabla_{\mathbf{r}} = \mathbf{i}\frac{\partial}{\partial r_x} + \mathbf{j}\frac{\partial}{\partial r_y} + \mathbf{k}\frac{\partial}{\partial r_z} \tag{2.171}$$

and

$$\mathcal{H}_{\text{Kepler}} = \frac{p^2}{2\mu^*} - \frac{\mu\mu^*}{r}. \tag{2.172}$$

Here, as previously, $\mu = \mathcal{G}(m_1 + m_2)$. Also, $p = |\mathbf{p}|$ and $r = |\mathbf{r}|$. The new quantity

$$\mu^* = \frac{m_1m_2}{m_1 + m_2} \tag{2.173}$$

is called the *reduced mass* of the system. The quantity $\mathcal{H}_{\text{Kepler}}$ is referred to as the *Hamiltonian* of the Kepler (i.e., two-body) problem. We have now replaced the three, coupled, second-order differential equations by an analogous system of six, coupled, first-order differential equations given by

$$\dot{\mathbf{r}} = \frac{\mathbf{p}}{\mu^*} \qquad \text{and} \qquad \dot{\mathbf{p}} = -\frac{\mu\mu^*}{r^3}\mathbf{r}. \tag{2.174}$$

If we compare Eq. (2.172) with Eq. (2.28) it is clear that $\mathcal{H}_{\text{Kepler}} = \mu^*C$. In our new formulation $\mathcal{H}_{\text{Kepler}}$ is the sum of the kinetic and potential energy of the system and so equals the total energy, a constant of the system.

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In general, any system of equations that can be written in the form

$$\frac{\mathrm{d}q_i}{\mathrm{d}t} = +\frac{\partial \mathcal{H}}{\partial p_i}, \qquad \frac{\mathrm{d}p_i}{\mathrm{d}t} = -\frac{\partial \mathcal{H}}{\partial q_i} \qquad (i = 1, 2, \ldots, n), \tag{2.175}$$

where $\mathcal{H} = \mathcal{H}(q_i, p_i, t)$, is said to be a *Hamiltonian* system of *order* $2n$, or, equivalently, of n *degrees of freedom*. The function $\mathcal{H}$ is called the *Hamiltonian* of the system. The variables q_i ($i = 1, 2, \ldots, n$) are called the *coordinates* while the p_i ($i = 1, 2, \ldots, n$) are called the *momenta*, which are said to be *conjugate* to the q_i. Although in our case above the q_i are the coordinates of the body and the p_i are the components of the momentum, there is no requirement that the description of the variables has to conform to their role. Note that $\mathcal{H}$ is determined only up to an arbitrary additive constant, since $\mathcal{H} + k$, where k is a constant, also satisfies Eq. (2.175).

At first glance it may seem perverse to express the equations of motion in this form. However, the properties of such systems make coordinate transformations easier to carry out and this will be particularly useful when we investigate resonant phenomena. In addition, such transformations may simplify the problem and lead to a solution. It is beyond the scope of this book to provide a rigorous introduction to the theory of Hamiltonian systems; indeed only a few results are required for the subjects we cover. Elementary treatments are given by Brouwer & Clemence (1961) and Roy (1988).

We have already demonstrated that the variables $\mathbf{r}$ and $\mathbf{p}$ form a conjugate set. Unfortunately, it is easy to show that the same is not true of the orbital elements a, e, I, Ω, ω, and f that can be derived from them. However, certain functions of the orbital elements can form conjugate sets. Here we mention two such sets: the *Delaunay variables* and the *Poincaré variables*.

The Delaunay variables are defined by

$$l = M, \qquad g = \omega, \qquad h = \Omega,$$
$$L = \mu^* \sqrt{\mu a}, \quad G = \mu^* \sqrt{\mu a(1 - e^2)}, \quad H = \mu^* \sqrt{\mu a(1 - e^2)} \cos I, \tag{2.176}$$

where l, g, and h are the coordinates and L, G, and H are the corresponding conjugate momenta. The Hamiltonian for the two-body problem expressed in terms of the Delaunay variables is

$$\mathcal{H} = -\frac{\mu^2 \mu^{*3}}{2L^2}. \tag{2.177}$$

Since $\mathcal{H}$ is only a function of L it implies that g and h are constants (clearly the case from Sect. 2.8) and L, G, and H are also constants (also clear since $L = L(a)$, G is the angular momentum, and H is the vertical component of the angular momentum vector).

The variation of l is given by

$$\frac{\mathrm{d}l}{\mathrm{d}t} = +\frac{\partial \mathcal{H}}{\partial L} = \frac{\mu^2 \mu^{*3}}{L^3} = \sqrt{\frac{\mu}{a^3}}, \tag{2.178}$$

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which is to be expected since $\mathrm{d}l/\mathrm{d}t = \mathrm{d}n(t-\tau)/\mathrm{d}t = n$.

The Poincaré variables are defined by

$$\begin{aligned} \lambda &= M + \omega + \Omega, & \Lambda &= \mu^* \sqrt{\mu a}, \\ \gamma &= -\omega - \Omega, & \Gamma &= \mu^* \sqrt{\mu a}\left(1 - \sqrt{1-e^2}\right), \\ z &= -\Omega, & Z &= \mu^* \sqrt{\mu a (1-e^2)}(1 - \cos I), \end{aligned} \tag{2.179}$$

where λ, γ, and z are the coordinates and Λ, Γ, and Z are the corresponding conjugate momenta. The angle $\lambda = M + \varpi$ is the mean longitude. The Poincaré variables can be derived from the Delaunay variables by the transformation

$$\begin{aligned} \Lambda &= L, & \Gamma &= L - G, & Z &= G - H, \\ \lambda &= l + g + h, & \gamma &= -g - h, & z &= -h. \end{aligned} \tag{2.180}$$

Note that the following relationship between the two sets of variables holds:

$$\Lambda\lambda + \Gamma\gamma + Zz = Ll + Gg + Hh. \tag{2.181}$$

This is an example of what is called a *contact transformation* (see, for example, Brouwer & Clemence 1961) and such a transformation always preserves the canonical nature of the equations without change in the Hamiltonian. The Hamiltonian for the two-body problem expressed in terms of the Poincaré variables is

$$\mathcal{H} = -\frac{\mu^2 \mu^{*3}}{2\Lambda^2}. \tag{2.182}$$

There is no requirement for a new set of variables to consist of coordinates and momenta: Mixed systems are also possible. Consider the variables

$$\xi = \sqrt{2\Gamma}\cos\gamma, \quad \eta = \sqrt{2\Gamma}\sin\gamma, \quad p = \sqrt{2Z}\cos z, \quad q = \sqrt{2Z}\sin z. \tag{2.183}$$

This is another example of a contact transformation. In this case the variables ξ and η are called the *eccentric variables* whereas p and q are called the *oblique variables*. When e and I are small these are the variables h, k, p, and q used in the discussion of secular perturbations in Chapter 7.

Exercise Questions

2.1 Consider a two-body problem in which the force of attraction is proportional to the radial separation of the two bodies (rather than the square of its inverse). Show that the motion of one mass with respect to the other is a centred ellipse.

2.2 Use the semi-major axes listed in Table A.2 to determine the average interval between times of orbital conjunction between the Earth and Mars. Show that for fixed orbits the *minimum* distance between Earth and Mars varies by almost a factor of two. What is the approximate interval between successive "very close" oppositions? Use the values of a_0, e_0, ϖ_0, and λ_0 given in Table A.2 (but not their variation given in Table A.3) and numerical solutions of Kepler's equation to determine the orbital motions of Earth and Mars over the period 1985–2002. Neglecting the relative orbital inclinations of the Earth and Mars, show that the closest opposition during this period occurred in September 1988 and the farthest in February 1995, and determine the minimum distances at these times.

2.3 A test particle approaches a planet of mass M and radius R from infinity with speed v_∞ and an impact parameter p. Use the particle's energy and angular momentum with respect to the planet to derive expressions for the semi-major axis and eccentricity of the hyperbolic orbit followed by the test particle about M, and for the pericentre distance r_0. Show that the eccentricity may be written $e = 1 + 2v_\infty^2/v_0^2$, where v_0 is the escape velocity at r_0. Use the expression for the true anomaly corresponding to the asymptote of the hyperbola ($r \to \infty$) to show that the overall deflection of the test particle's orbit after it leaves the vicinity of the planet, ψ, is given by $\sin(\psi/2) = e^{-1}$. Given that r_0 must be greater than R to avoid a physical collision, calculate the maximum deflection angles for (i) a spacecraft skimming Jupiter, with $v_\infty = 10$ km s^{-1}, and (ii) the *Cassini* orbiter skimming Saturn's large moon Titan, at $v_\infty = 5$ km s^{-1}.

2.4 Consider the relative motion form of the two-body problem. Explain why the vector $\mathbf{e} = -(\mathbf{h} \times \mathbf{v})/(\mathcal{G}(m_1 + m_2)) - \hat{\mathbf{r}}$ lies in the orbital plane, where $\mathbf{h}$ is the angular momentum per unit mass, $\mathbf{v}$ is the velocity, and $\hat{\mathbf{r}}$ is the unit radial vector. Show that $\mathbf{e}$ is a vector constant of the motion. By expressing the dot product $\mathbf{e} \cdot \mathbf{r}$ in two different forms, where $\mathbf{r}$ is the position vector, show that the orbit is an ellipse of eccentricity $e = |\mathbf{e}|$ and longitude of pericentre ϖ defined as the angle that $\mathbf{e}$ makes with the reference $\hat{\mathbf{x}}$ direction.

2.5 In July 1994 the *Galileo* spacecraft, on its way to Jupiter, observed the fragments of Comet Shoemaker-Levy 9 impacting the planet. The first impact occurred on 16 July 1994 when the spacecraft was moving along a near-zero inclination heliocentric orbit with orbital elements $a = 3.137$ AU, $e = 0.690$, and $\varpi = 82.2^\circ$, where ϖ is the longitude of perihelion. The mean anomaly of the spacecraft was 45.7° at an epoch 322 days prior to the first impact. Find the spacecraft's eccentric anomaly, true anomaly, radial distance from the Sun, and true longitude (in the standard reference system) on 16 July 1994. At the time of the first impact Jupiter had a true longitude of 225.4 degrees and a heliocentric distance of 5.417 AU. Assuming that Jupiter's orbit has zero inclination, show that in terms of longitude coverage on the surface of the planet, approximately 28% of the unilluminated side was observable by the *Galileo* spacecraft.

2.6 There have been suggestions that Comet P/Swift-Tuttle could have a close approach to the Earth in AD 2126. The Earth was at the perihelion of its orbit on 4 January 1993 at 3h UT. Assuming that the inclination of the Earth's orbit is zero and taking its semi-major axis, eccentricity, and longitude of perihelion to have the fixed values 1 AU, 0.0167, and 102.996° respectively, calculate its position vector 48,799.375 days later at noon on 14 August 2126. Observations suggest that Comet P/Swift-Tuttle was at the perihelion of its orbit on 12 December 1992 at 21h 23m UT with the following orbital elements: $a = 26.35441$ AU, $e = 0.96362$, $I = 113.408^\circ$, $\omega = 152.979^\circ$, $\Omega = 139.430^\circ$, where a is the semi-major axis, e is the eccentricity, I is the inclination, ω is the argument of perihelion, and Ω is the longitude of ascending node. Calculate the position vector of the comet 48,821.609 days later at noon on 14 August 2126. Use your answer from the first part of the question to calculate the separation of the comet and the Earth (in AU) at this date in the future. What are the major sources of error in determining this separation?